

PMATH 465

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1 Manifolds

1.1 Coordinate Charts

The concept of manifolds starts with the notion of coordinate chart.

Definition 1.1. A (coordinate) chart on a set X is a subset $U \subseteq X$ together with a bijection

$$\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$$

onto an open subset $\varphi(U)$ of $\mathbb{R}^n$.

This allows us to parametrise points $x \in U$ by n coordinates $\varphi(x) = (x_1, \dots, x_n)$.

Example 1.2. $X = \mathbb{R}^n, U = X, \varphi = \text{id}_{\mathbb{R}^n}, x \mapsto x$. Also, for any $U \subseteq \mathbb{R}^n$ open, $\varphi = \text{id}_{\mathbb{R}^n} : U \rightarrow U, x \mapsto x$.

Example 1.3. $U \subseteq \mathbb{R}^n$ open, $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ map. We can set

$$\begin{aligned} X &= \{(x, y) \in U \times \mathbb{R}^m : y = f(x)\} \subset \mathbb{R}^n \times \mathbb{R}^m \\ &= \text{graph of } f. \end{aligned}$$

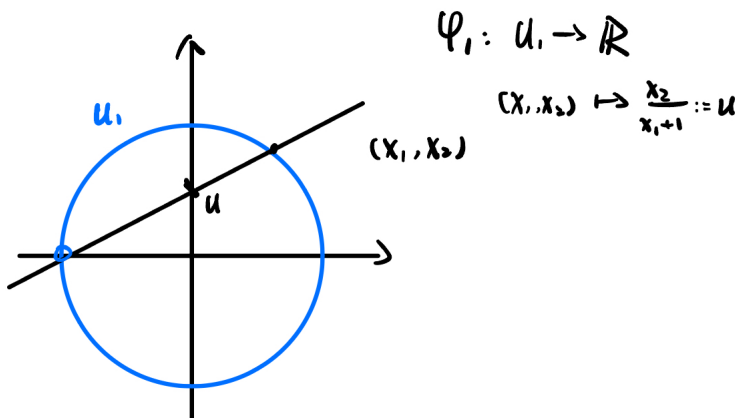
Then we can set

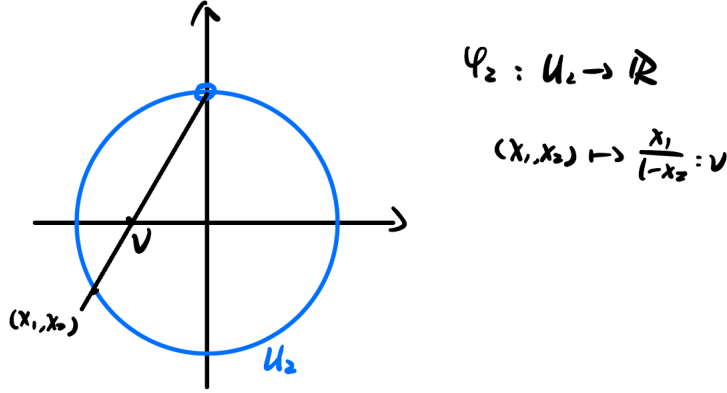
$$\begin{aligned} \varphi : X &\rightarrow U = \varphi(U) \subset \mathbb{R}^n, \text{ open} \\ (x, y) &\mapsto x \end{aligned}$$

is a coordinate chart on X .

Example 1.4.

$$\begin{aligned} X &= S^1 = \{(x_1, x_2) \in \mathbb{R}^2, x_1^2 + x_2^2 = 1\} \\ &= \text{unit circle in } \mathbb{R}^2 \end{aligned}$$





Example 1.5. $X = \{\text{straight lines in } \mathbb{R}^2\}$. For non-vertical lines, we can write straight lines in the form $x_2 = mx_1 + b$ uniquely. Let $U_1 := \{\text{non-vertical lines}\} = \{x_2 = mx_1 + b : m, b \in \mathbb{R}\}$. Let

$$\varphi_1 : U_1 \rightarrow \mathbb{R}^2 = \varphi_1(U_1) \text{ via } x_2 = mx_1 + b \mapsto (m, b).$$

Similarly, let $U_2 = \{\text{non-horizontal lines}\} = \{x_1 = \tilde{m}x_2 + \tilde{b} : \tilde{m}, \tilde{b} \in \mathbb{R}\}$, and let

$$\varphi_2 : U_2 \rightarrow \mathbb{R}^2 \text{ via } x_1 = \tilde{m}x_2 + \tilde{b} \mapsto (\tilde{m}, \tilde{b}).$$

As we have seen, a set X can be covered by more than one chart. We want these charts to satisfy some consistency conditions.

Definition 1.6. A n -dimensional (smooth) atlas on X is a collection of coordinate charts

$$\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$$

such that

1. X is covered by the $\{U_\alpha\}_{\alpha \in A}$ (i.e., $X = \bigcup_{\alpha \in A} U_\alpha$). Also, $\varphi_\alpha : U_\alpha \subset X \rightarrow \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n$ is open, and φ_α bijection.
2. $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n, \forall \alpha, \beta \in A$.
3. The map

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$$

is smooth.

Remark 1.7. 1. Since the indices are running through all $\alpha, \beta \in A$, we in fact have that $\varphi_\alpha \circ \varphi_\beta^{-1}$ is a diffeomorphism.

2. The maps $\varphi_\alpha \circ \varphi_\beta^{-1}$ are “change of coordinate” maps.

Example 1.8. 1. $X = \mathbb{R}^n, \varphi : X \rightarrow X, x \mapsto x$. Then $\{(X, \varphi)\}$ is n -dimensional atlas for $\mathbb{R}^n$. Or, let $X = U \subset \mathbb{R}^n$ open, $\varphi : \text{id}_{\mathbb{R}^n} : U \rightarrow U, x \mapsto x$. Since there is only one chart, $\{(X, \varphi)\}$ is an n -dimensional smooth atlas on U .

2. In general, if X can be covered by only one chart $\varphi : X \rightarrow \varphi(X) \subset \mathbb{R}^n$, then $\{(X, \varphi)\}$ is an n -dimensional smooth atlas. For instance, graphs of functions admit a smooth atlas.

3. $X = S^1$, $\varphi_1 : U_1 \rightarrow \mathbb{R}$, $\varphi_2 : U_2 \rightarrow \mathbb{R}$ is a 1-dimensional smooth atlas since

- (a) $S^1 = U_1 \cup U_2$
- (b) $\varphi_1(U_1 \cap U_2) = \mathbb{R} \setminus \{x_2 = 1\}$ open in $\mathbb{R}$
- (c) $\varphi_2(U_1 \cap U_2) = \mathbb{R} \setminus \{x_1 = -1\}$ open in $\mathbb{R}$
- (d) $\varphi_1 \circ \varphi_2^{-1} : \varphi_2(U_1 \cap U_2) \rightarrow \varphi_1(U_1 \cap U_2)$ via $v \mapsto \frac{v-1}{(v+1)^2} = u$ is a diffeomorphism.

Example 1.9. Other atlas: $X = S^1 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$ [insert pic here]

Example 1.10. $X = \{\text{lines in } \mathbb{R}^2\}$, then

$$U_1 = \{\text{non-vertical lines}\}, \varphi_1 : U_1 \rightarrow \mathbb{R}^2 \text{ via } \{x_2 = mx_1 + b\} \mapsto (m, b)$$

$$U_2 = \{\text{non-horizontal lines}\}, \varphi_2 : U_2 \rightarrow \mathbb{R}^2 \text{ via } \{x_1 = \bar{m}x_2 + \bar{b}\} \mapsto (\bar{m}, \bar{b})$$

Example 1.11. Real Projective space $\mathbb{RP}^n$.

$$\begin{aligned} \mathbb{RP}^n &:= \{1\text{-dimensional subspaces of } \mathbb{R}^{n+1} \text{ (viewed as a vector space)}\} \\ &= \{\text{lines through } 0 \text{ in } \mathbb{R}^{n+1}\} \end{aligned}$$

So, if $L \in \mathbb{RP}^n$, then L is a line through 0 in $\mathbb{R}^{n+1}$, all we need to completely describe L as a direction vector $0 \neq v \in \mathbb{R}^{n+1}$.

$$\implies L = L_v := \{\text{lines through } 0 \text{ in } \mathbb{R}^{n+1} \text{ with direction } v\}$$

Notice that $L_v = L_{\lambda v}, \forall \lambda \in \mathbb{R} \setminus \{0\} = \mathbb{R}^\times$. Atlas: $U_i := \{L_v : v_i \neq 0\}$, then $\forall L \in U_i, L_v = L \frac{1}{v_i} v$, which we called the preferred direction vector. So, we can choose the direction vector $\frac{1}{v_i} v$ for L which is such that its i -coordinate is 1. BUT, note that L has only one direction vector whose i th coordinate is 1. We thus get a bijection

$$\varphi_i : U_i = \{L_v : v_i \neq 0\} \rightarrow \mathbb{R}^n = \{x \in \mathbb{R}^{n+1} : x_i = 1\} \quad \text{via } L_v \mapsto \frac{1}{v_i} v$$

with $\varphi_i(U_i) = \mathbb{R}^n$ is open. Also, $\mathbb{RP}^n = \bigcup_{i=1}^{n+1} U_i$, and $\varphi_i(U_i \cap U_j) \subset \mathbb{R}^n$ is open in $\mathbb{R}^n$. Lastly,

$$\varphi_i \circ \varphi_j^{-1} : \varphi_j(U_i \cap U_j) \rightarrow \varphi_i(U_i \cap U_j)$$

via $x \mapsto \frac{1}{x_i} x$ is smooth.

1.2 Manifolds

Definition 1.12. A n -dimensional (smooth) manifold is a set that admits a n -dimensional smooth atlas. BUT a good definition should not depend on the choice of atlas.

Definition 1.13. Two atlases $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ and $\{(V_i, \psi_i)\}_{i \in I}$ are compatible if their union is again an atlas.

Remark 1.14. In particular, this means that $\varphi_\alpha \circ \psi_i^{-1}$ is a diffeomorphism, $\forall \alpha, i$. Also, compatibility is an equivalence relation (exercise!).

Definition 1.15. A smooth/differentiable structure on a set X is an equivalence class of atlases.

Definition 1.16. A smooth manifold is a set together with a smooth atlas structure.

NOTE:

1. By definition, to show that a set X is a manifold, you just need to construct one smooth atlas.
2. Given a smooth structure on a set X , the maximal atlas consists of the union of ALL the charts in the atlases in the equivalence class.

Many books define a smooth manifold as a topological space that can be covered by charts that are “smoothly compatible” (i.e., satisfy condition 3 of the definition of atlases). In fact, the atlas $\{(U_\alpha, \varphi_\alpha)\}$ induces a natural topology on X .

Definition 1.17. Let X be a set. A topology on X is a collection τ of subsets of X that satisfies

1. $\emptyset, X \in \tau$.
2. τ is closed under countable unions.
3. τ is closed under finite intersection.

The elements of τ are called open. $F \subseteq X$ is said to be closed if F^c is open.

Remark 1.18. $\emptyset, X$ are open and closed.

Example 1.19. 1. $\tau = \{\emptyset, X\}$ is a topology on X .

2. $X = \mathbb{R}^n$, $\tau = \{\text{unions of open balls}\} = \text{standard topology on } \mathbb{R}^n$.

3. Let τ be a topology on X and $A \subset X$, we can set

$$\tau_A := \{A \cap U : U \in \tau\} = \text{relative topology on } A.$$

One can check that τ_A is a topology on A where closed sets are of the form $A \cap F$ with F closed in X .

Suppose that M is a smooth n -dimensional manifold. Let $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ be a smooth atlas in the smooth structure of M .

Definition 1.20. A subset $V \subseteq M$ is said to be open if

$$\varphi_\alpha(V \cap U_\alpha) \subset \mathbb{R}^n \text{ is open, } \forall \alpha$$

Proposition 1.21. We have U_α is open in M , for all $\alpha \in A$ and these open sets define a topology in M .

Proof. U_α is open since $\varphi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$ is open for all β by condition 2 of atlas. Also,

1. $\varphi(\emptyset \cap U_\alpha) = \varphi_\alpha(\emptyset) = \emptyset$ open in $\mathbb{R}^n$.
2. $\varphi_\alpha(M \cap U_\alpha) = \varphi_\alpha(U_\alpha)$ is open in $\mathbb{R}^n$.
3. Let $\{V_i\}_{i \in I}$ be a collection of open sets in M so that $\varphi_\alpha(V_i \cap U_\alpha)$ is open in $\mathbb{R}^n, \forall i, \alpha$. Then

$$\varphi_\alpha((\bigcup_{i \in I} V_i) \cap U_\alpha) = \varphi_\alpha(\bigcup_{i \in I} (V_i \cap U_\alpha)) = \bigcup_{i \in I} \varphi_\alpha(V_i \cap U_\alpha) \text{ is open}$$

4. Similarly, let $\{V_i\}_{i=1}^m$ be open sets in M , then

$$\varphi_\alpha((\bigcup_{i=1}^m V_i) \cap U_\alpha) = \varphi_\alpha(\bigcup_{i=1}^m (V_i \cap U_\alpha)) = \bigcup_{i=1}^m \varphi_\alpha(V_i \cap U_\alpha) \text{ is open}$$

□

Proposition 1.22. The bijection $\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n$ are homeomorphisms with respect to this topology on M (and the standard topology in $\mathbb{R}^n$).

Proof. Need to check that φ_α and φ_α^{-1} are continuous.

- φ_α^{-1} is continuous: notice that $\varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n \rightarrow U_\alpha \subset M$. Let $V \subset M$ be open, then

$$(\varphi_\alpha^{-1})^{-1}(V \cap U_\alpha) = \varphi_\alpha(V \cap U_\alpha) \text{ is open in } \mathbb{R}^n$$

therefore φ_α^{-1} is continuous.

- φ_α is continuous: notice that $\varphi_\alpha : U_\alpha \subset M \rightarrow \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n$. Let $W \subset \mathbb{R}^n$ be open. Need to check that $\varphi_\alpha^{-1}(W)$ is open in M , that is, $\varphi_\beta(\varphi_\alpha^{-1}(W) \cap U_\beta)$ is open in $\mathbb{R}^n$, for all β . BUT,

$$\varphi_\beta(\varphi_\alpha^{-1}(W) \cap U_\beta) = \varphi_\beta \circ \varphi_\alpha^{-1}(W \cap \varphi_\alpha(U_\beta \cap U_\alpha))$$

which is open since $U_\beta \circ \varphi_\alpha^{-1}$ is a homeomorphism, therefore $\varphi_\alpha^{-1}(W)$ is open in $U_\alpha \subset M$.

□

In practice, we want the topology to be Hausdorff and also second countable.

Definition 1.23. (X, τ) is called Hausdorff if $\forall p \neq q \in X$, there exists $U, V \in \tau$ with $p \in U, q \in V$, $U \cap V = \emptyset$.

Example 1.24. 1. $X = \mathbb{R}^n, \tau = \tau_d$ is Hausdorff.

2. $X, \tau = \{\emptyset, X\}$ not Hausdorff.

Definition 1.25. A basis $\mathcal{B}$ of τ is a collection of open sets such that

0. Every $U \in \tau$ is a union of elements in $\mathcal{B}$.

1. $\forall x \in X, \exists B \in \mathcal{B}$ such that $x \in B$,

2. If $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subset B_1 \cap B_2$.

The elements of the basis $\mathcal{B}$ are the “building blocks” of τ : one can show that one can recover τ from $\mathcal{B}$.

Example 1.26. 1. $X = \mathbb{R}^n, \tau = \tau_d$. Then $\mathcal{B}$ can be taken to be the set of all open balls.

2. Given an atlas $\{U_\alpha, \varphi_\alpha\}_{\alpha \in A}$ of the manifold M , we have that $\mathcal{B} = \{U_\alpha\}_{\alpha \in A}$ is a basis of the natural topology on M .

Definition 1.27. (X, τ) is called second countable if it admits a countable basis.

Since we want the natural topology on M to also be Hausdorff and second countable, two additional conditions shall be added to atlas:

4 Countably many of the sets U_α covers M , that is, $M = \bigcup_{j \in \mathbb{N}} U_{\alpha_j}$ for some $\alpha_j \in A$. (gives second countability)

5 $\forall p, q \in M$ with $p \neq q$, either $p, q \in U_\alpha$ for some α or there exists α, β with $U_\alpha \cap U_\beta = \emptyset$ and $p \in U_\alpha, q \in U_\beta$. (gives Hausdorff)

Lemma 1.28. Condition (4) implies τ is second countable, and Condition (5) implies τ is Hausdorff.

Proof. Suppose (5) holds. Let $p \neq q \in M$ be arbitrary. If $\exists \alpha, \beta \in A$ with $U_\alpha \cap U_\beta = \emptyset$ and $p \in U_\alpha, q \in U_\beta$, we are done. Otherwise, $\exists \alpha \in A$ with $p, q \in U_\alpha$. Note that

$$\varphi_\alpha : U_\alpha \xrightarrow{\text{homeo}} \varphi_\alpha(U_\alpha) \subset \mathbb{R}^n$$

and $\mathbb{R}^n$ is Hausdorff with respect to the standard topology. Let $d = \text{dist}(\varphi_\alpha(p), \varphi_\alpha(q))$. Set $V_p = B_{d/3}(\varphi_\alpha(p))$, $V_q = B_{d/3}(\varphi_\alpha(q))$. Then V_p, V_q are disjoint open sets with $\varphi_\alpha(p) \in V_p, \varphi_\alpha(q) \in V_q$. Since φ_α is a homeomorphism,

$$\varphi_\alpha^{-1}(V_p), \varphi_\alpha^{-1}(V_q) \in \tau$$

with $p \in \varphi_\alpha^{-1}(V_p), q \in \varphi_\alpha^{-1}(V_q)$, and $\varphi_\alpha^{-1}(V_p) \cap \varphi_\alpha^{-1}(V_q) = \emptyset$.

Suppose that (4) holds. Then there are countably many of the U_α 's cover M . Say $M = \bigcup_{m \in \mathbb{N}} U_{\alpha_m}$. Then, $\varphi_{\alpha_m}(U_{\alpha_m})$ is an open set in $\mathbb{R}^n$, which is second countable. So, we can pick a countable basis of $\varphi_{\alpha_m}(U_{\alpha_m})$. Then the union of the preimages under $\varphi_{\alpha_m}^{-1}$ of these basis is a second countable basis of τ (since the union of bases is again a basis). $\square$

Recall: $M = \mathbb{RP}^n = \{\text{lines through } 0 \text{ in } \mathbb{R}^{n+1}\}$. Then for all $L \in \mathbb{RP}^n$, $L = L_v = L_{\lambda v}, \forall \lambda \in \mathbb{R}^\times$, where $v = (v_1, \dots, v_{n+1})$ is a vector along L . Set $U_i := \{L_v : v_i \neq 0\}$. Then

$$\varphi_i : U_i \rightarrow \mathbb{R}^n = \{x \in \mathbb{R}^{n+1} : x_i = 1\} \quad \text{via} \quad v \mapsto \frac{1}{v_i} v.$$

Therefore $\{(U_i, \varphi_i)\}_{i=1}^{n+1}$ is a smooth atlas on $\mathbb{RP}^n$, gives a natural topology on $\mathbb{RP}^n$. Also, condition (4) is satisfied since the atlas has a finite number of charts. That is, $\mathbb{RP}^n$ is second countable. (5) is left as an exercise.

Definition 1.29 (abstract definition of a manifold). Let (X, τ) be a topological space, and assume it is Hausdorff and second countable. We say (X, τ) is a topological n -dimensional manifold if $\exists n \in \mathbb{N}$ such that $\forall p \in X$, there exists open neighborhood U of p in τ and a homeomorphism

$$\varphi : U \rightarrow \varphi(U) \subset \mathbb{R}^n \quad \text{open.}$$

The map φ is called a coordinate chart.

Definition 1.30. An atlas for a topological n -dimensional manifold is a collection of coordinate charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ such that $X = \bigcup_{\alpha \in A} U_\alpha$.

Definition 1.31. An atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ on (X, τ) is called smooth if

$$\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$$

is smooth, $\forall \alpha, \beta \in A$.

Definition 1.32. A smooth structure on a topological n -dimensional manifold (X, τ) is an equivalence class of smooth atlases in which case (X, τ) is called a smooth n -dimensional manifold.

Remark 1.33. The original second condition is satisfied since φ is a homeomorphism, so it maps open sets to open sets.

Example 1.34. Manifold that is not Hausdorff: line with double origin. Fix $a \neq b \in \mathbb{R}$.

$$L := (\mathbb{R} \times \{a\}) \cup (\mathbb{R} \times \{b\}) / \sim$$

where $(x, a) \sim (y, b)$ if $x = y \neq 0$.

$$\frac{[(0, a)]}{\bullet} \frac{[(0, b)]}{\bullet}$$

Let $U_1 := L \setminus \{[(0, b)]\}$, $\varphi_1 : U_1 \rightarrow \mathbb{R} = \mathbb{R} \times \{a\}$ via $[(x, a)] \rightarrow x$. Then it is a bijection.

Let $U_2 := L \setminus \{[(0, a)]\}$, $\varphi_2 : U_2 \rightarrow \mathbb{R} = \mathbb{R} \times \{b\}$ via $[(y, b)] \rightarrow y$. Then it is a bijection.

1. $U_1 \cup U_2 = L$
2. $U_1 \cap U_2 = L \setminus \{[(0, a)], [(0, b)]\}$, then $\varphi_1(U_1 \cap U_2) = \mathbb{R} \setminus \{0\} = \varphi_2(U_1 \cap U_2)$ open in $\mathbb{R}$.
3. $\varphi_1 \circ \varphi_2^{-1} : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ via $y \mapsto \underbrace{[(y, b)]}_{=[(y, a)]} \mapsto y$. Therefore $\varphi_1 \circ \varphi_2^{-1} = \text{id}_{\mathbb{R} \setminus \{0\}}$, which is a diffeomorphism.

Therefore $\{(U_1, \varphi_1), (U_2, \varphi_2)\}$ is an atlas. Hence L is a smooth 1-dimensional manifold. Also, the natural topology on L induced by this atlas has open sets $V \subset L$ if

$$\varphi_1(V \cap U_1), \varphi_2(V \cap U_2)$$

are open in $\mathbb{R}$. Away from $\{(0, a), (0, b)\}$, this is just the standard topology on $\mathbb{R}$, which is Hausdorff. But, any open set V containing $[(0, a)]$ will intersect any open set containing $[(0, b)]$ because both contain points

$$(-\varepsilon, 0) \cup (0, \varepsilon)$$

for some $\varepsilon > 0$. Thus, the topology is not Hausdorff.

Example 1.35. A manifold that is not second countable: An uncountable disjoint union of copies of $\mathbb{R}$.

Theorem 1.36. Let $F : U \subset \mathbb{R}^{n+m} \text{ open} \rightarrow \mathbb{R}^m$ is a smooth function and let $c \in \mathbb{R}^m$. Consider

$$F^{-1}(c) := \{x \in U : f(x) = c\} \subset U$$

Suppose that $\forall a \in F^{-1}(c)$, the derivative

$$DF_a : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$$

is surjective. Then, $F^{-1}(c)$ has the structure of a smooth n -dimensional manifold that is Hausdorff and second countable.

Example 1.37. $S^n := \{R^{n+1} : x_1^2 + \dots + x_{n+1}^2 = 1\}$ = n -sphere in $\mathbb{R}^{n+1}$. Let

$$F : \mathbb{R}^{n+1} \rightarrow \mathbb{R} \quad \text{via} \quad (x_1, \dots, x_{n+1}) \rightarrow x_1^2 + \dots + x_{n+1}^2.$$

Then, $S^n = F^{-1}(1)$. Also,

$$DF_x = \begin{bmatrix} \frac{\partial F}{\partial x_1} & \dots & \frac{\partial F}{\partial x_{n+1}} \end{bmatrix} = \begin{bmatrix} 2x_1 & \dots & 2x_{n+1} \end{bmatrix}$$

So, for all $x \in S^n$, DF_x is surjective (as $DF_x = 0 \iff x = 0$, but there is no $x = 0$ on n -sphere). Therefore S^n is a smooth manifold of dimension $(n+1) - 1 = n$.

Example 1.38. $M_{n \times n}(\mathbb{R}) = \{n \times n \text{ real matrices}\} = \mathbb{R}^{n^2}$. Then $M_{n \times n}$ is a smooth n^2 -dimensional manifold with coordinates a_{ij} with $i, j = 1, \dots, n$. Let $c \in \mathbb{R}^\times$ be fixed, define

$$X_c := \{A \in M_{n \times n}(\mathbb{R}) : \det A = c\}$$

For instance, when $c = 1$, then

$$X_1 := \text{SL}(n, \mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) : \det A = 1\}$$

the special linear group. Consider

$$F = \det : \mathbb{R}^{n^2} = M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}^1 \quad \text{via} \quad A \mapsto \det A$$

Then $X_c = F^{-1}(c)$. We want to check that $DF_A : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ is surjective. Notice

$$DF_A = \left(\frac{\partial F}{\partial a_{ij}} \right) = \begin{bmatrix} \frac{\partial F}{\partial a_{11}} & \dots & \frac{\partial F}{\partial a_{1n}} & \frac{\partial F}{\partial a_{21}} & \dots & \frac{\partial F}{\partial a_{nn}} \end{bmatrix}$$

Then, DF_A is surjective $\iff$ it is not the zero matrix $\iff \frac{\partial F}{\partial a_{ij}} \neq 0$ for some $i_0, j_0 \in [n]$. Note that $F(A) = \det A$, and

$$\frac{\partial \det A}{\partial a_{ij}} = \text{adj}(A)_{ji}$$

where $\text{adj}(A)$ = adjoint matrix of A = (cofactor matrix of A)^T, as $\det A = \sum_j a_{ij}(\text{adj} A)_{ij}^T$. Then

$$\frac{\partial \det A}{\partial a_{ij}} = (\text{adj} A)_{ij}^T = (\text{adj} A)_{ji}$$

If $(\text{adj} A)_{ji} = 0, \forall i, j = 1, \dots, n$, then $\det A = 0$. So such i_0, j_0 exists and therefore DF_A is surjective. Thus, $F^{-1}(c)$ is a smooth $(n^2 - 1)$ -dimensional manifold. In particular, $\text{SL}(n, \mathbb{R})$ is a smooth $(n^2 - 1)$ -dimensional manifold.

1.3 Smooth maps between manifolds

Suppose that M, N are smooth manifolds with dimensions m, n respectively. Consider a map

$$F : M \rightarrow N$$

Definition 1.39. A map $F : M \rightarrow N$ is called smooth if $\forall x \in M$, and chart $(U_\alpha, \varphi_\alpha)$ in M with $x \in U_\alpha$ and (V_i, ψ_i) in N with $F(x) \in V_i$, the set $F^{-1}(V_i)$ is open in M and the composition

$$\psi_i \circ F \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap F^{-1}(V_i)) \subset \mathbb{R}^m \rightarrow \psi_i(V_i \cap F(U_\alpha)) \subset \mathbb{R}^n$$

is a C^∞ function.

Remark 1.40. 1. The condition $F^{-1}(V_i)$ is open in M ensures that F is continuous as a map from M to N with respect to the natural topology we have on M and N .

2. The charts $(U_\alpha, \varphi_\alpha)$ and (V_i, ψ_i) can only be chosen within the smooth structures of M and N .
3. It is enough to check smoothness of F for a choice of atlas in the smooth structures of M and N . For example, suppose $(\tilde{U}_\beta, \tilde{\varphi}_\beta)$ is another chart in M with $x \in \tilde{U}_\beta$. Then,

$$\varphi_\alpha \circ \tilde{\varphi}_\beta^{-1}$$

is smooth. So

$$\psi_i \circ F \circ \tilde{\varphi}_\beta^{-1} = (\psi_i \circ F \circ \varphi_\alpha^{-1}) \circ (\varphi_\alpha \circ \tilde{\varphi}_\beta^{-1})$$

is a composition of two smooth functions, and therefore it is smooth. Similarly, we get the same thing with the ψ_i 's.

The natural notion of equivalence between manifolds is:

Definition 1.41. A diffeomorphism $F : M \rightarrow N$ is a smooth map with a smooth inverse.

Remark 1.42. If M and N are diffeomorphic, then they have the same dimension because any diffeomorphism $F : M \rightarrow N$ is such that

$$\psi_i \circ F \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap F^{-1}(V_i)) \subset \mathbb{R}^m \rightarrow \psi_i(V_i \cap F(U_\alpha)) \subset \mathbb{R}^n$$

is a diffeomorphism $\implies m = n$.

Example 1.43. 1. $M = N = \mathbb{R}^n$ and $F = \text{id}_{\mathbb{R}^n}$, then F is a diffeomorphism. (Here, $\varphi = \psi = \text{id}_{\mathbb{R}^n}$ so that $\psi \circ F \circ \varphi^{-1} = \text{id}_{\mathbb{R}^n}$.)

2. For any chart $\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha)$ on M , φ_α is a diffeomorphism, because

$$\text{id}_{\varphi_\alpha(U_\alpha)} \circ \varphi_\alpha \circ \varphi_\alpha^{-1} = \text{id}_{\varphi_\alpha(U_\alpha)}$$

is a diffeomorphism (and $\varphi_\alpha^{-1}(V_i) = \varphi_\alpha^{-1}(\varphi_\alpha(U_\alpha)) = U_\alpha$ open).

3. $\text{id}_M : M \rightarrow M$, where they have the same smooth structure, is a diffeomorphism ($\varphi_\alpha \circ \text{id} \circ \varphi_\alpha^{-1} = \text{id}_{\varphi_\alpha(U_\alpha)}$ is a diffeomorphism for any charts $(U_\alpha, \varphi_\alpha)$ on M).

4. Suppose that M is the graph of a function

$$f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m, U \text{ open.}$$

Then, M is a smooth manifold with atlas

$$\{U, \varphi : M = \{(x, y) \in U \times \mathbb{R}^m : y = f(x)\} \rightarrow U \text{ via } (x, y) \mapsto x\}$$

By example 2, φ is a diffeomorphism from M to U . Therefore, the graphs of functions are diffeomorphic to their domain. Notice that there is no condition on f for this to be true (does not have to be smooth), e.g., $f : \mathbb{R} \rightarrow \mathbb{R}$ via $x \mapsto |x|$. As a submanifold of $\mathbb{R}^2$ with the induced topology from $\mathbb{R}^2$, M has a singularity at the origin. But, as an abstract manifold it is diffeomorphic to $\mathbb{R}$, and is thus smooth.

5. $M = \mathbb{R}/\mathbb{Z}$ where $a \sim b$ if $b - a \in \mathbb{Z}, \forall a, b \in \mathbb{R}$. Consider

$$p : \mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$$

via

$$a \mapsto [a]_\sim$$

Let $U_0 = M \setminus \{[0]_\sim\}$, $\varphi_0 = p|_{(0,1)}^{-1} : U_0 \rightarrow \varphi_0(U_0) = (0, 1)$ via $[a]_\sim \rightarrow a$. Then, φ_0 is a bijection because any equivalent class $[a]_\sim \in U_0$ can be represented by $a \in (0, 1)$.

Let $U_1 = M \setminus \{[\frac{1}{2}]_\sim\} \implies [0]_\sim \in U_1$, $\varphi_1 = p|_{(-\frac{1}{2}, \frac{1}{2})}^{-1} : U_1 \rightarrow \varphi_1(U_1) = (-\frac{1}{2}, \frac{1}{2}) \subset \mathbb{R}$ via $[a]_\sim \rightarrow a$. Again, φ_1 is a bijection because any $[a]_\sim \in U_1$ can be represented by $a \in (-\frac{1}{2}, \frac{1}{2})$. Then,

- $M = U_0 \cup U_1$
- $\varphi_0(U_0 \cap U_1) = (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ open, and $\varphi_1(U_0 \cap U_1) = (-\frac{1}{2}, 0) \cup (0, \frac{1}{2})$ open.

- $\varphi_0 \circ \varphi_1^{-1} : \varphi_1(U_0 \cap U_1) = (-\frac{1}{2}, 0) \cup (0, \frac{1}{2}) \rightarrow \varphi_0(U_0 \cap U_1) = (0, \frac{1}{2}) \cup (\frac{1}{2}, 1)$ via

$$x \mapsto \begin{cases} x+1, & x \in (-\frac{1}{2}, 0) \\ x, & x \in (0, \frac{1}{2}) \end{cases} \text{ is smooth.}$$

Similarly,

$$\varphi_1 \circ \varphi_0^{-1} = \begin{cases} x, & x \in (0, \frac{1}{2}) \\ x-1, & x \in (\frac{1}{2}, 1) \end{cases}$$

is also smooth.

Therefore, $M = \mathbb{R}/\mathbb{Z}$ is a smooth manifold.

Claim: M is diffeomorphic to $S^1 = \{(x_1, x_2) : x_1^2 + x_2^2 = 1\}$.

Claim. Let $G : M = \mathbb{R}/\mathbb{Z} \rightarrow S^1$ via $[x]_{\sim} \rightarrow (\cos(2\pi x), \sin(2\pi x))$. Then, G is a bijection. $\square$

Lemma 1.44. If F is smooth, then F is continuous with respect to the natural topology on M and N .

Proof. Exercise! $\square$

Definition 1.45 (Lee's). $F : M \rightarrow N$ is smooth if $\forall x \in M$ and any chart $(U_\alpha, \varphi_\alpha)$ in M with $x \in U_\alpha$ and (V_i, ψ_i) in N , with $F(x) \in V_i$ and $F(U_\alpha) \subset V_i$, we have

$$\psi_i \circ F \circ \varphi_\alpha^{-1} \text{ is } C^\infty.$$

Lemma 1.46. F smooth $\implies F$ continuous.

Remark 1.47. Both definitions are equivalent.

Proof. Lee's def $\implies$ def: by Lemma 1.46, F is continuous and therefore $F^{-1}(V_i)$ is open.
def $\implies$ Lee's def: $F^{-1}(V_i)$ open, then $U_\alpha \cap F^{-1}(V_i)$ open. Set

$$\tilde{U}_\alpha = U_\alpha \cap F^{-1}(V_i)$$

Then $F(\tilde{U}_\alpha) \subset V_i$, and $\tilde{\varphi}_\alpha = \varphi_\alpha|_{\tilde{U}_\alpha}$. $\square$

2 Tangent vectors and cotangent vectors

2.1 Smooth functions on manifolds

Let M be a manifold and consider a map

$$f : M \rightarrow \mathbb{R}$$

Note that $\mathbb{R}$ is a manifold with atlas $\{\mathbb{R}, \text{id}_{\mathbb{R}}\}$ (so that the induced topology on $\mathbb{R}$ is the standard topology). Then, f is smooth if $\forall x \in M$ and $(U_\alpha, \varphi_\alpha)$ in M with $x \in U_\alpha$, and $(\mathbb{R}, \text{id}_{\mathbb{R}})$ in $\mathbb{R}$, $f^{-1}(\mathbb{R}) = M$ open, and

$$\text{id}_{\mathbb{R}} \circ f \circ \varphi_\alpha^{-1}$$

is C^∞ on $\varphi_\alpha(f^{-1}(\mathbb{R}) \cap U_\alpha) = \varphi_\alpha(U_\alpha)$. In this case, we get a simpler definition for smooth scalar functions:

Definition 2.1. A scalar function $f : M \rightarrow \mathbb{R}$ is smooth if $\forall x \in M$ and chart $(U_\alpha, \varphi_\alpha)$ in M with $x \in U_\alpha$,

$$f \circ \varphi_\alpha^{-1}$$

is C^∞ on $\varphi_\alpha(U_\alpha) \subset \mathbb{R}^m$ (where $m = \dim M$).

Notation: M^m means that M is a m -dimensional manifold.

$C^\infty(M) = \{\text{set of all smooth scalar functions in } M\}$.

Remark 2.2. Note that $C^\infty(M)$ is a real vector space.

Recall: Compact subspace of a topological space (X, τ) ,

- If $X = \mathbb{R}^n, \tau = \text{standard topology}$, then compact subsets of $(\mathbb{R}^n, \tau)$ corresponds to closed and bounded subsets.

Definition 2.3. A subset $K \subset X$ is compact if for any collection $\{U_\alpha\}_{\alpha \in A}$ of open sets in X with

$$K \subset \bigcup_{\alpha \in A} U_\alpha,$$

there exists $\alpha_1, \dots, \alpha_k$ with $K \subset U_{\alpha_1} \cup \dots \cup U_{\alpha_k}$.

Proposition 2.4. τ Hausdorff. Then K compact $\implies K$ closed.

Proof. Suppose that K is compact. We need to show that $X \setminus K$ is open in X . We do this by proving that $X \setminus K$ is a union of open sets. It is enough to show that $\forall x \in X \setminus K$, there exists open $U_x \ni x$ with $U_x \subset X \setminus K$ such that

$$X \setminus K = \bigcup_{x \in X \setminus K} U_x$$

Let $x \in X \setminus K$. Then, $\forall y \in K$, we have $x \neq y$. So there exists open V_y and U_y such that

$$x \in V_y, y \in U_y, V_y \cap U_y = \emptyset.$$

Then $K \subset \bigcup_{y \in K} U_y$. Since K is compact, there exists $y_1, \dots, y_s$ such that $K \subset U_{y_1} \cup \dots \cup U_{y_s}$. Set $U_x := V_{y_1} \cap \dots \cap V_{y_s}$ open in X with $x \in U_x$ and $U_x \cap K = \emptyset$ (since $V_{y_i} \cap U_{y_i} = \emptyset$). Therefore $U_x \subset X \setminus K$. $\square$

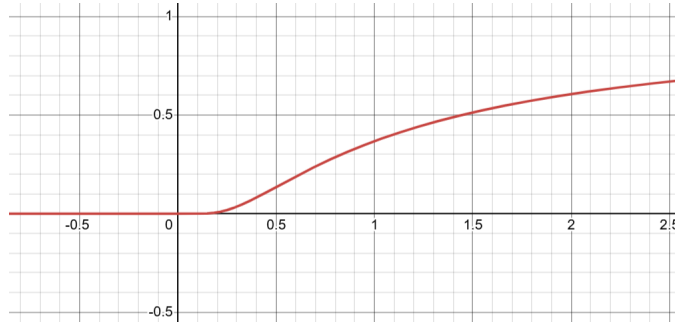
Fact: Let (X, τ) and (X', τ') be two topological spaces. Consider a map $F : X \rightarrow X'$:

1. If F is continuous and $K \subset X$ is compact, then $F(K) \subset X'$ is compact
2. If F is a homeomorphism and $K' \subset X'$ is compact, then $F^{-1}(K') \subset X$ is compact.

We want to show that smooth functions always exist on a manifold M . We first introduce the notion of a bump function:

Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

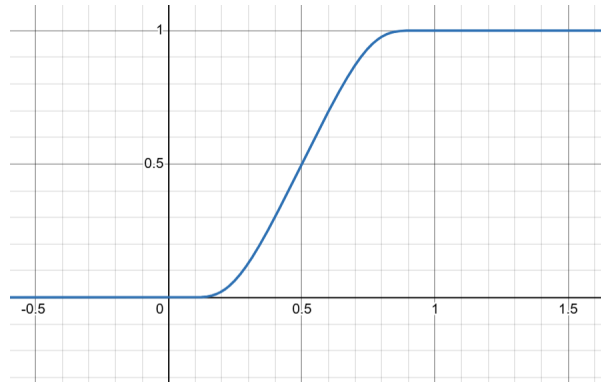
$$f(t) = \begin{cases} e^{-\frac{1}{t}} & , t > 0 \\ 0 & , t \leq 0 \end{cases}$$



Then, f is C^∞ . Set

$$g(t) = \frac{f(t)}{f(t) + f(1-t)}, t \in \mathbb{R}.$$

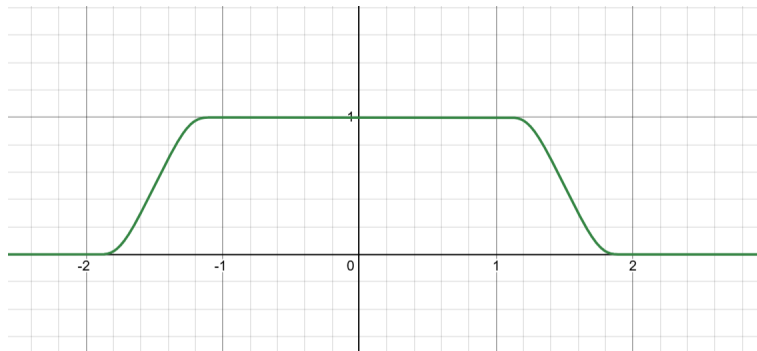
Then, $g : \mathbb{R} \rightarrow \mathbb{R}$ is smooth. Also, g is identically 1 when $t \geq 1$ and identically 0 when $t \leq 0$.



Next,

$$h(t) = g(2+t)g(2-t), t \in \mathbb{R}$$

Again, $h : \mathbb{R} \rightarrow \mathbb{R}$ is smooth and vanishes for $|t| \geq 2$, and $h(t) = 1$ if $|t| \leq 1$.



In $\mathbb{R}^n$, we can define $H : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$H(x_1, \dots, x_n) := h(x_1)h(x_2) \cdots h(x_n).$$

Then $H \equiv 1$ if $|x| \leq 1$ (sup norm) and 0 outside an open ball $B_r(0)$. Let M^n be a manifold and (U, φ) be a chart on M . Let H be the function we designed. We can assume, after possibly composing H is a contraction of $\mathbb{R}^n$, that $B_r(0) \subset \varphi(U) \subset \mathbb{R}^n$. Set

$$f : M \rightarrow \mathbb{R} \quad \text{via} \quad x \mapsto f(x) = \begin{cases} H \circ \varphi(x) & , x \in U \\ 0 & , x \notin U \end{cases}$$

Then, $\forall x \in U$, $f \circ \varphi^{-1} = H : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is C^∞ , and for points outside U , then $f \equiv 0$, and therefore C^∞ .

Note that f is not constant since H is not constant. Thus, there are infinitely many smooth functions on M .

2.2 Cotangent spaces

Let M^n be a manifold, $a \in M$, (U, φ) be a chart with $a \in U$ and $\varphi(x) = (x_1, \dots, x_n) \in \mathbb{R}^n$. For all $f \in C^\infty(M)$, we can define " $\frac{\partial f}{\partial x_i}$ ":

$$\frac{\partial f}{\partial x_i}|_a := \frac{\partial}{\partial x_i}(f \circ \varphi^{-1})|_{\varphi(a)}$$

Remark 2.5. Given an open cover $\{V_\alpha\}_{\alpha \in A}$ of a manifold M , there exists collection of smooth functions $\{f_\alpha : M \rightarrow \mathbb{R}\}$ such that

1. $f_\alpha \geq 0$ (that is, $f_\alpha(x) \geq 0, \forall x \in M$)
2. $\text{supp}(f_\alpha) := \overline{\{x \in M : f_\alpha(x) \neq 0\}} \subset V_\alpha$.
3. $\{\text{supp}(f_\alpha) : \alpha \in A\}$ is locally finite (that is, $\forall x \in M, f_\alpha(x) \neq 0$ for only a finite number of α 's).
4. $\sum_{\alpha \in A} f_\alpha(x) = 1, \forall x \in M$.

Definition 2.6. A collection of functions that satisfies these properties is called a partition of unity subordinate to the open cover $\{V_\alpha\}_{\alpha \in A}$.

Theorem 2.7. Given any open cover $\{V_\alpha\}_{\alpha \in A}$, there exists a partition of unity subordinate to it.

Remark 2.8. The partitions of unity are useful since they are used to define global objects on M from local objects on open set U_α with $\bigcup U_\alpha = M$.

Remark 2.9. Given a partition of unity $\{f_\alpha\}_{\alpha \in A}$ of $\{V_\alpha\}_{\alpha \in A}$, we have

1. $f_\alpha \equiv 0$ on $M \setminus V_\alpha$ by property 2,
2. $0 \leq f_\alpha(x) \leq 1, \forall \alpha \in A, x \in M$ by property 1 and 4.

Recall: Let (X, τ) be a topological space.

1. $\forall A \subset X$, we define $\overline{A} :=$ smallest closed set containing A =closure of A . In particular, $A \subset \overline{A}$, and $A = \overline{A}$ iff it is closed.

2. A subset $K \subset X$ is compact if for any open cover $\{U_i\}_{i \in I}$ of K ,

$$K \subset U_{i_1} \cup \cdots \cup U_{i_m}$$

for some $i_1, \dots, i_m \in I$. Heine-Borel: $\mathbb{R}^n$ with standard topology, compact $\iff$ closed and bounded. If τ is Hausdorff, then compact $\implies$ closed. For any τ , if K is compact and $F \subset K$ is closed in K , then F is compact.

3. If (X', τ') is another topological space, and $f : X \rightarrow X'$ is continuous, then $f(K)$ is compact for any $K \subset X$ compact. If f is a homeomorphism, then $f^{-1}(K')$ is compact for any $K' \subset X'$ compact.

Proposition 2.10. Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of M . Then there exists a collection of smooth functions $\{f_i\}_{i \in I}$ on M such that

1. $f_i \geq 0$,
2. $\forall x \in M, f_i(x) \neq 0$ for only finitely many i 's.
3. $\sum_{i \in I} f_i \equiv 1$,
4. $\text{supp}(f_i) \subset V_{\alpha(i)}$ for some $\alpha(i) \in A, \forall i \in I$.

The theorem then follows from this: $\forall \alpha \in A$, set

$$\tilde{f}_\alpha = \sum_{i: \alpha(i) = \alpha} f_i$$

Therefore,

$$\text{supp}(\tilde{f}_\alpha) \subset V_\alpha \text{ since } \text{supp}(f_i) \subset V_\alpha, \text{ when } \alpha(i) = \alpha.$$

Then, $\forall x \in M, \tilde{f}_\alpha \geq 0$ and $\tilde{f}_\alpha(x) \neq 0$ for only finitely many α 's. Set $f(x) := \sum_{\alpha \in A} \tilde{f}_\alpha(x)$, then $f : M \rightarrow \mathbb{R}$ is smooth. Set $f_\alpha = \frac{\tilde{f}_\alpha}{f}$, then $\sum_{\alpha \in A} f_\alpha = 1$.

Proof. Start with open cover $\{V_\alpha\}_{\alpha \in A}$ of M . We have the following:

1. $\forall x \in M$, there exists open neighborhood U of x in M with $\overline{U}$ compact.

Proof. $\forall x \in M$, there exists a coordinate chart $\varphi : x \in W \rightarrow \varphi(W) \subset \mathbb{R}^n$ open. Then, since $\varphi(W)$ is open in $\mathbb{R}^n$, and $\varphi(x) \in \varphi(W)$, there exists open ball $B_\varepsilon(\varphi(x)) \subset \varphi(W)$ with $\varepsilon > 0$. Then $B_{\frac{\varepsilon}{2}}(\varphi(x)) \subset \varphi(W)$ and $\overline{B_{\frac{\varepsilon}{2}}(\varphi(x))} \subset \varphi(W)$. Set $U = \varphi^{-1}(B_{\frac{\varepsilon}{2}}(\varphi(x)))$. Then, $x \in U$ and

$$\overline{U} = \varphi^{-1}(\overline{B_{\frac{\varepsilon}{2}}(\varphi(x))}) \text{ is compact since } \varphi \text{ is a homeomorphism}$$

□

2. $\overline{U}$ is closed.

3. The topology on M is second countable, so there exists a countable basis $\mathcal{B}$ of the topology. $\forall x \in M$, there exists an open neighborhood U of x with $\overline{U}$ compact therefore $\overline{U}$ closed. Also, there exists $U_j \in \mathcal{B}$ with $x \in \overline{U_j}$ and $U_j \subset U$. Therefore $\overline{U_j} \subset \overline{U}$, $\overline{U}$ is compact, so $\overline{U_j}$ is compact.

Then we get: M has a countable basis $\mathcal{B} = \{U_j\}_{j \in \mathbb{N}}$ such that $\overline{U_j}$ is compact, $\forall j \in \mathbb{N}$. Define $G_1 := U_1$, then $\overline{G_1} \subset \bigcup_{j=1}^{\infty} U_j$, $\overline{G_1}$ compact. Let $k > 0$ be the smallest integer such that

$$\overline{G_1} = U_1 \cup \dots \cup U_k$$

(which exists by compactness of $\overline{G_1}$).

Set $G_2 := \bigcup_{j=1}^k U_j$. Then $G_1 \subset G_2$ and $\overline{G_2} = \bigcup_{j=1}^k \overline{U_j}$ is compact. By doing it inductively, we get

$$\overline{G_j} \subset G_{j+1}, M = \bigcup_{j=1}^{\infty} G_j.$$

Also, $\overline{G_j} \setminus G_{j-1}$ is compact, and $G_{j+2} \setminus \overline{G_{j-1}}$ is open (exercise!). Let $x \in M$. Then $x \in V_\alpha$ and $x \in \overline{G_j} \setminus G_{j-1} \subset G_{j+1} \setminus \overline{G_{j-1}}$. Then $x \in V_\alpha \cap (G_{j+2} \setminus \overline{G_{j-1}})$ open, there exists a smooth function f on M whose support is contained in $V_\alpha \cap (G_{j+1} \setminus G_{j-1}) \subset V_\alpha$. $\square$

Application of partition of unity:

Let M be a manifold and $A \subset M$ a closed subset. Also, let $U \subset M$ be an open set such that $A \subset U$.

Proposition 2.11. There exists a smooth $\psi : M \rightarrow \mathbb{R}$ such that $\psi \equiv 1$ on A and $\text{supp} \psi \subset U$, where

$$\text{supp} \psi := \overline{\{x \in M : \psi(x) \neq 0\}}.$$

In particular, $\psi \equiv 0$ on $M \setminus U$.

Proof. Let $U_0 = U$ and $U_1 = M \setminus A$. Since A is closed, U_1 is open in U . Also, since $A \subset U$,

$$M = \underbrace{A}_{\subset U=U_0} \cup \underbrace{(M \setminus A)}_{=U_1} = U_0 \cup U_1$$

Therefore, $\{U_0, U_1\}$ is an open cover of M . Then, there exists a partition of unity subordinate to $\{U_0, U_1\}$: that is, there exists smooth functions $f_0, f_1 : M \rightarrow \mathbb{R}$ such that

- $\text{supp} f_i \subset U_i, i = 0, 1 \implies f_i \equiv 0$ on $M \setminus U_i$,
- $f_0 + f_1 = 1$.

Let $\psi := f_0 : M \rightarrow \mathbb{R}$. Then, $\text{supp} \psi = \text{supp} f_0 \subset U_0 = U$. Also, $\forall x \in A$, we know that $f_0(x) + f_1(x) = 1$, and if $x \in A$, then

$$x \notin M \setminus A = U_1 \implies x \in M \setminus U_1 \implies f_1(x) = 0 \implies f_0(x) = 1$$

$\square$

2.3 The derivative of a function

Let M be a manifold, and $a \in M$. Given a smooth function $f : M \rightarrow \mathbb{R}$, how can we define the derivative of f at a ?

Suppose that $M = \mathbb{R}^n$ with coordinates $(x_1, \dots, x_n)$. Then, we have

$$\frac{\partial f}{\partial x_i}(a) := \lim_{h \rightarrow 0} \frac{f(a + he_i) - f(a)}{h} = \text{partial of } f \text{ at } a \text{ w.r.t. } x_i,$$

where $\{e_1, \dots, e_n\}$ is the standard basis. Then:

$$Df(a) := \left[\frac{\partial f}{\partial x_1}(a) \quad \dots \quad \frac{\partial f}{\partial x_n}(a) \right] = \text{derivative of } f \text{ at } a.$$

Note that if $g : M \rightarrow \mathbb{R}$ is another smooth function, then

$$\frac{\partial}{\partial x_i}(f - g)(a) = \frac{\partial f}{\partial x_i}(a) - \frac{\partial g}{\partial x_i}(a),$$

so that

$$Df(a) = Dg(a) \iff D(f - g)(a) = 0 = [0 \quad \dots \quad 0].$$

Let us now consider a general manifold M . Let us choose a chart (U, φ) of M with $a \in U$. Then

$$f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$$

is smooth. Let $\varphi(x) = (x_1(x), \dots, x_n(x))$. Then, the following partials exist at $\varphi(a)$:

$$\frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\varphi(a)), \forall i = 1, \dots, n.$$

But, the values $\frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\varphi(a))$ may depend on the choice of chart (U, φ) ! Nonetheless, if $\frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\varphi(a)) = 0, \forall i = 1, \dots, n$, then

$$\frac{\partial}{\partial y_j}(f \circ \psi^{-1})(\psi(a)) = 0, \forall j = 1, \dots, n$$

for any other chart (V, ψ) with $a \in V$ and $\psi(x) = (y_1(x), \dots, y_n(x))$.

Proof. Say $g = f \circ \psi^{-1}$ and $h = f \circ \varphi^{-1}$. Then

$$h = f \circ \psi^{-1} = \underbrace{(f \circ \varphi^{-1})}_{=g} \circ \underbrace{(\varphi \circ \psi^{-1})}_{\text{diffeo}}.$$

Then, by the Chain Rule,

$$Dh(\psi(x)) = Dg(\varphi(x)) \cdot D(\varphi \circ \psi^{-1})(\psi(x))$$

since $h(\psi(x)) = g((\varphi \circ \psi^{-1})(\psi(x))) = g(\varphi(x))$. Note that $F = \varphi \circ \psi^{-1} : \psi(V \cap U) \subset \mathbb{R}^n \rightarrow \varphi(U \cap V) \subset \mathbb{R}^n$ is a diffeomorphism. So, F is smooth and F^{-1} is smooth. Also,

$$F \circ F^{-1} = \text{id}_{\varphi(U \cap V)} \quad \text{and} \quad F^{-1} \circ F = \text{id}_{\psi(V \cap U)}$$

Therefore

$$DF \cdot DF^{-1} = \text{Did}_{\varphi(U \cap V)} = I_{n \times n}, DF^{-1} \cdot DF = \text{Did}_{\psi(V \cap U)} = I_{n \times n}.$$

Therefore, DF is invertible on $\psi(U \cap V)$. So:

$$Dh(\psi(x)) = 0 \iff Dg(\varphi(x)) = 0$$

□

We get the following well-defined notion:

Definition 2.12. The smooth function $f : M \rightarrow \mathbb{R}$ has derivative equal to 0 at $a \in M$ if $D(f \circ \psi^{-1})(\varphi(a)) = 0$ for any chart (U, φ) of M with $a \in U$. Let

$$Z_a := \{f \in C^\infty(M) \mid f \text{ has zero derivative at } a\} \subset C^\infty(M)$$

Definition 2.13. Let $a \in M$. The cotangent space T_a^*M of M at a is defined as:

$$T_a^*M := C^\infty(M)/Z_a$$

Notice that $C^\infty(M)$ is a vector space over $\mathbb{R}$ since

$$f + cg \in C^\infty(M), \forall f, g \in C^\infty(M), c \in \mathbb{R}$$

because given any chart (U, φ) ,

$$(f + cg) \circ \psi^{-1} = (f \circ \psi^{-1}) + c(g \circ \psi^{-1}) \text{ is } C^\infty.$$

Similarly, one check: if $f, g \in Z_a, c \in \mathbb{R}$, then $(f + cg) \in Z_a$ (exercise!). Therefore, we conclude

- Z_a is a subspace of $C^\infty(M)$
- $C^\infty(M)/Z_a$ is a quotient vector space over $\mathbb{R}$.

Also, $\forall f \in C^\infty(M)$,

$$(df)_a = [f]_{Z_a} = \text{equivalence class of } f \text{ in } T_a^*M.$$

In particular, if $f, g \in C^\infty(M)$, then

$$(df)_a = (dg)_a \iff (f - g) \in Z_a \iff g = f + h \text{ for some } h \in Z_a.$$

Definition 2.14. For all $f \in C^\infty(M)$, $(df)_a$ is called the derivative of f at a .

Remark 2.15. By the above, two functions $f, g \in C^\infty(M)$ have the same derivative at a iff $(f - g)$ has derivative 0 at a .

Definition 2.16. The tangent space T_aM to M at a is

$$T_aM := (T_a^*M)^*$$

Proposition 2.17. Let M be an n -dimensional manifold and $a \in M$. Then:

1. T_a^*M is an n -dimensional real vector space,
2. If (U, φ) is an arbitrary coordinate chart on M with $a \in U$, and $\varphi = (x_1, \dots, x_n)$, then

$$\{(dx_1)_a, \dots, (dx_n)_a\}$$

is a basis for T_a^*M ,

3. $\forall f \in C^\infty(M)$,

$$(df)_a = \sum_{i=1}^n \underbrace{\left(\frac{\partial}{\partial x_i} (f \circ \varphi^{-1})(\varphi(a)) \right)}_{\in \mathbb{R}} (dx_i)_a$$

Remark 2.18. If $f, g \in C^\infty(M)$ such that $f = g$ on open subset $W \subset M$, with $a \in W$, then

$$X_a(f) = X_a(g)$$

Proof. Let $h = f - g$. If $h \equiv 0$ on W , then

$$X_a(h) = 0$$

Therefore we have

$$\begin{aligned}\psi h &\equiv h \text{ on } M \\ \psi &\equiv 1 \text{ on } A = \text{supp} h \\ \psi &\equiv 0 \text{ on } M \setminus \{0\}.\end{aligned}$$

So $X_a(h) = X_a(\psi h) = 0$. □

M manifold, $a \in M$,

$$Z_a = \{f \in C^\infty(M) : \text{all derivatives of } f \text{ vanishes at } a\}$$

Then

$$T_a^* M = C^\infty(M)/Z_a = \{(f)_{Z_a} =: (df)_a : f \in C^\infty(M)\}$$

If $(a, \varphi = (x_1, \dots, x_n))$ is a chart with $a \in M$, then

$$T_a^* M = \text{span}_{\mathbb{R}}\{(dx_1)_a, \dots, (dx_n)_a\} = \text{cotangent space at } a$$

and

$$(df)_a := \sum_{i=1}^n \underbrace{\frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\varphi(a))}_{\text{coeff of } (df)_a \text{ wrt the basis}} (dx_i)_a$$

Here, $f \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R}$ smooth, and

$$T_a M = (T_a^* M)^* = \text{tangent space at } a$$

and

$$T_a M = \text{span}_{\mathbb{R}}\{(\frac{\partial}{\partial x_1})_a, \dots, (\frac{\partial}{\partial x_n})_a\}$$

where $\{(\frac{\partial}{\partial x_1})_a, \dots, (\frac{\partial}{\partial x_n})_a\}$ is the dual basis of $\{(dx_i)_a : i = 1, \dots, n\}$ so that

$$(\frac{\partial}{\partial x_i})_a((dx_j)_a) = \delta_{i,j}.$$

In fact,

$$T_a M = \text{Der}_a(M) = (\text{derivatives at } a) = \text{span}_{\mathbb{R}}\{\frac{\partial}{\partial x_1}|_a, \dots, \frac{\partial}{\partial x_n}|_a\}$$

where

$$\frac{\partial}{\partial x_i}|_a(f) := \frac{\partial}{\partial x_i}(f \circ \varphi^{-1})(\varphi(a))$$

change of variable: Suppose that (V, ψ) is another chart with $\psi = (y_1, \dots, y_n)$. Then,

$$T_a M = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial y_1}|_a, \dots, \frac{\partial}{\partial y_n}|_a \right\}$$

with

$$\frac{\partial}{\partial y_j}|_a(f) := \frac{\partial(f \circ \varphi^{-1})}{\partial y_j}(\psi(a)), j = 1, \dots, n$$

Note that $\varphi \circ \psi^{-1}(y_1, \dots, y_n) = (x_1, \dots, x_n)$, therefore $(x_1, \dots, x_n) = (x_1(y_1, \dots, y_n), \dots, x_n(y_1, \dots, y_n))$. So

$$\begin{aligned} \frac{\partial}{\partial y_j}|_a(f) &= \frac{\partial}{\partial y_j}(f \circ \psi^{-1})(\psi(a)) \\ &= \frac{\partial}{\partial y_j}((f \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}))(\psi(a)) \\ &= \frac{\partial}{\partial x_1}(f \circ \varphi^{-1})(\varphi(a)) \frac{\partial x_1}{\partial y_j}(\psi(a)) + \dots + \frac{\partial}{\partial x_n}(f \circ \varphi^{-1})(\varphi(a)) \frac{\partial x_n}{\partial y_j}(\psi(a)), \text{ by Chain Rule} \\ &\implies \frac{\partial}{\partial y_j}|_a(f) = \sum_{i=1}^n \frac{\partial x_i}{\partial y_j}(\psi(a)) \frac{\partial}{\partial x_i}|_a(f) \\ &\implies \frac{\partial}{\partial y_j}|_a = \sum_{i=1}^n \frac{\partial x_i}{\partial y_j}(\psi(a)) \frac{\partial}{\partial x_i}|_a, \forall i, j = 1, \dots, n \end{aligned}$$

and

$$\left(\frac{\partial x_i}{\partial y_j}(\psi(a)) \right)_{i,j=1}^n$$

is the change of basis matrix from $\left\{ \frac{\partial}{\partial x_i}|_a : i = 1, \dots, n \right\}$ to $\left\{ \frac{\partial}{\partial y_j}|_a : j = 1, \dots, n \right\}$.

Note that

$$\left(\frac{\partial x_i}{\partial y_j}(\psi(a)) \right)_{i,j=1}^n = \text{Jac}(\varphi \circ \psi^{-1})(\psi(a))$$

Similarly, the change of basis matrix from $\{(dx_i)_a : i = 1, \dots, n\}$ to $\{(dy_j)_a : j = 1, \dots, n\}$ is given by the Jacobian of $\varphi \circ \psi^{-1}$ at $\psi(a)$ (Exercise!).

2.4 Derivative of Smooth Maps

Definition 2.19. Let M and N be two smooth manifolds. Let $F : M \rightarrow N$ be a smooth map. Let $a \in M$. The derivative at a of F is the homomorphism of tangent spaces

$$DF_a = F_{*,a} : T_a M \rightarrow T_{F(a)} N$$

such that $\forall X_a \in T_a M$, $DF_a(X_a) \in T_{F(a)} N$ is given by

$$DF_a(X_a)(f) := X_a(f \circ F), \forall f \in C^\infty(M)$$

Remark 2.20. $X_a : C^\infty(M) \rightarrow \mathbb{R}$ is $\mathbb{R}$ -linear and satisfies Leibniz. Since $f : N \rightarrow \mathbb{R}$ and $F : M \rightarrow N$ are both smooth,

$$f \circ F : M \rightarrow \mathbb{R}$$

is smooth. So, $X_a(f \circ F)$ is well-defined. Also, the $\mathbb{R}$ -linear and Leibniz for $DF_a(X_a)$ follows from X_a (Exercise!). Concretely, suppose that $(U, \varphi = (x_1, \dots, x_n))$ is a chart on M with $a \in U$ and $(V, \psi = (y_1, \dots, y_m))$ is a chart on N with $F(a) \in V$. Then, for all $f \in C^\infty(N)$,

$$\begin{aligned} DF_a\left(\frac{\partial}{\partial x_i}\bigg|_a\right)(f) &:= \frac{\partial}{\partial x_i}\bigg|_a(f \circ F) \\ &= \frac{\partial}{\partial x_i}(f \circ F \circ \varphi^{-1})(\varphi(a)) \end{aligned}$$

$$\begin{aligned} DF_a\left(\frac{\partial}{\partial x_i}\bigg|_a\right)(f) &= \frac{\partial}{\partial x_i}(f \circ F \circ \varphi^{-1})(\varphi(a)) \\ &= \frac{\partial}{\partial x_i}((f \circ \psi^{-1}) \circ \underbrace{(\psi \circ F \circ \varphi^{-1})}_{\tilde{F}})(\varphi(a)) \\ &= \sum_{j=1}^m \frac{\partial}{\partial y_j}(f \circ \psi^{-1})(\psi(F(a))) \frac{\partial \tilde{F}}{\partial x_i}(\varphi(a)), \text{ by Chain Rule} \\ &= \sum_{j=1}^m \frac{\partial \tilde{F}}{\partial x_i}(\varphi(a)) \frac{\partial}{\partial y_j}\bigg|_a(f) \\ \implies DF_a\left(\frac{\partial}{\partial x_i}\bigg|_a\right) &= \sum_{j=1}^m \frac{\partial \tilde{F}_j}{\partial x_i}(\varphi(a)) \frac{\partial}{\partial y_j}\bigg|_a. \end{aligned}$$

Note that

$$\left(\frac{\partial \tilde{F}}{\partial x_i}\right)(\varphi(a))_{i=1, \dots, n, j=1, \dots, m} = \text{Jac}(\tilde{F})$$

is the matrix representation of DF_a with respect to the basis $\{\frac{\partial}{\partial x_i}\big|_a\}$ and $\{\frac{\partial}{\partial y_j}\big|_{F(a)}\}$ of T_aM and $T_{F(a)}N$ respectively.

Remark 2.21. 1. If F is a diffeomorphism so that F has a smooth inverse, then DF_a is an isomorphism, because then $\psi \circ F \circ \varphi^{-1}$ is a diffeomorphism, implying that $\text{Jac}(\psi \circ F \circ \varphi^{-1})(\varphi(a))$ is an invertible matrix.

2. We have

$$\frac{\partial}{\partial x_i}\bigg|_a = (\varphi^{-1})_{*, \varphi(a)}\left(\frac{\partial}{\partial x_i}\bigg|_{\varphi(a)}\right), \forall i = 1, \dots, n$$

$$[DF_a = F_{*, a} = \text{Derivative of } F \text{ at } a = \text{pushforward of } F \text{ at } a.]$$

Proof. $\varphi : U \subset M \rightarrow \varphi(U) \subset \mathbb{R}^n$, and $\varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow U \subset M$.

$$(\varphi^{-1})_{*, \varphi(a)}\left(\frac{\partial}{\partial x_i}\bigg|_{\varphi(a)}\right)(f) := \frac{\partial}{\partial x_i}\bigg|_{\varphi(a)}(f \circ \varphi^{-1}) =: \frac{\partial}{\partial x_i}\bigg|_a(f), \forall f \in C^\infty(M)$$

Note that $\varphi : U \subset M \rightarrow \varphi(U) \subset \mathbb{R}^n$ is a diffeomorphism. Therefore,

$$\varphi_{*,a} : T_a M \xrightarrow{\sim} T_{\varphi(a)} \mathbb{R}^n$$

is an isomorphism. □

second last description of $T_a M$: Let $\gamma : (-\varepsilon, \varepsilon) \subset \mathbb{R} \rightarrow M$ be a smooth map, $\varepsilon > 0$. Then image of γ in M is a smooth curve in M . Let $a = \gamma(0)$, set:

$$\gamma'(0) := \gamma_{*,a} \left(\frac{d}{dt} \Big|_{t=0} \right) = \text{velocity vector to } \gamma \text{ at } a.$$

Then

$$T_a M = \{\text{all velocity vectors to curves on } M \text{ passing through } a\}.$$

Remark 2.22. U is itself a manifold with chart $\varphi : U \rightarrow \varphi(U)$. So,

$$T_a U = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x_i} \Big|_a \right\} = T_a M$$

Example 2.23. $\text{GL}(n, \mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) : \det A \neq 0\} \subset M_{n \times n}(\mathbb{R})$ open. So $\forall A \in \text{GL}(n, \mathbb{R})$, $T_A \text{GL}(n, \mathbb{R}) = T_A M_{n \times n}(\mathbb{R})$.

Definition 2.24. Let $F : M \rightarrow N$ smooth map, and $a \in M$. The derivative (differential/pushforward map) of F at a is the map

$$DF_a = dF_a = F_{*,a} : T_a M \rightarrow T_{F(a)} N$$

defined by

$$DF_a(X_a)(f) := X_a(f \circ F)$$

$\forall X_a \in T_a M$ and $f \in C^\infty(M)$. Then, $DF_a(X_a)$ is the pushforward of X_a by F at a .

Proposition 2.25. $F : M \rightarrow N, G : N \rightarrow P$ smooth maps. Let $a \in M$,

- (a) $DF_a : T_a M \rightarrow T_{F(a)} N$ is $\mathbb{R}$ -linear.
- (b) $D(G \circ F)_a = DG_{F(a)} \circ DF_a : T_a M \rightarrow T_{G \circ F(a)} P$.
- (c) $D(\text{id}_M)_a = \text{id}_{T_a M}$.
- (d) If F is a diffeomorphism, then DF_a is an isomorphism of vector spaces, and

$$(DF_a)^{-1} = D(F^{-1})_{F(a)}$$

- (e) If M is connected, then $DF_a \equiv 0, \forall a \in M$ if and only if F is constant on M .

Remark 2.26. 1. $\frac{\partial}{\partial x_i} \Big|_a(f) := \frac{\partial}{\partial x_i} \Big|_{\varphi(a)}(f \circ \varphi^{-1}) = (\varphi^{-1})_{*,\varphi(a)} \left(\frac{\partial}{\partial x_i} \Big|_{\varphi(a)} \right)(f), \forall f \in C^\infty(M)$. So,

$$\frac{\partial}{\partial x_i} \Big|_a := (\varphi^{-1})_{*,\varphi(a)} \left(\frac{\partial}{\partial x_i} \Big|_{\varphi(a)} \right).$$

2. $F : M \rightarrow N$, $a \in M$, $(U, \varphi = (x_1, \dots, x_n))$ on M with $a \in U$, $(V, \psi = (y_1, \dots, y_m))$ on N with $F(a) \in V$. For all $X_a \in T_a M$, we have

$$DF_a(X - a) \in T_{F(a)}N = \text{span}_{\mathbb{R}}\left\{\frac{\partial}{\partial y_j}\bigg|_{F(a)}\right\}$$

Thus,

$$DF_a(X_a) = \sum_{j=1}^m c_j \frac{\partial}{\partial y_j}\bigg|_{F(a)} \text{ for some } c_1, \dots, c_m \in \mathbb{R}.$$

Since $\frac{\partial}{\partial y_j}\big|_{F(a)}(y_i) = \delta_{i,j}$, we have that

$$C_j = DF_a(X_a)(y_j)$$

So,

$$DF_a(X_a) = \sum_{j=1}^m DF_a(X_a)(y_j) \frac{\partial}{\partial y_j}\bigg|_{F(a)}.$$

2.5 Another Interpretation of $T_a M$:

Definition 2.27. A smooth curve on M is the image in M of a smooth map $\gamma : (-\varepsilon, \varepsilon) \rightarrow M, \varepsilon > 0$.

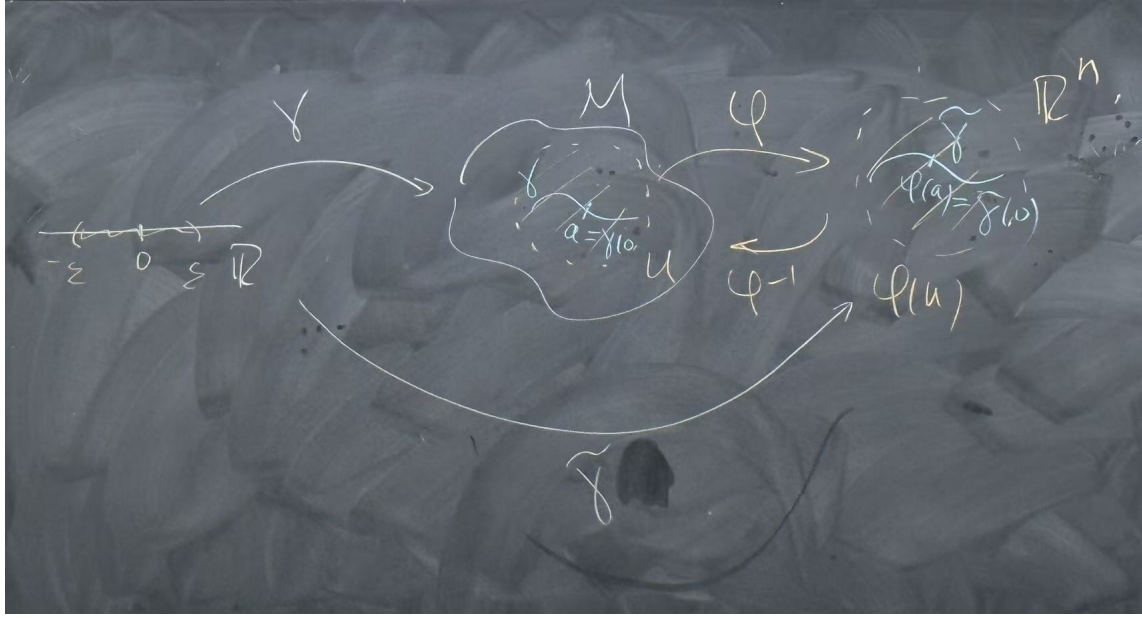
Let $a = \gamma(0) \in M$ and $(U, \varphi = (x_1, \dots, x_n))$ be a chart of M with $a \in U$. After possibly shrinking $(-\varepsilon, \varepsilon)$, we can assume that the image of γ is contained in U . Set

$$\tilde{\gamma} := \varphi \circ \gamma : (-\varepsilon, \varepsilon) \rightarrow \varphi(U) \subset \mathbb{R}^n$$

is a smooth curve in $\mathbb{R}^n$. In fact,

$$\gamma = \varphi^{-1} \circ \tilde{\gamma}$$

and any smooth curve in U is of this form.



Definition 2.28. The velocity vector of γ at $\gamma(t_0)$ is

$$\gamma'(t_0) := \gamma_{*,t_0} \left(\frac{d}{dt} \Big|_{t=t_0} \right), \forall t_0 \in (-\varepsilon, \varepsilon)$$

Note that t is the coordinate on $\mathbb{R}$ and $\frac{d}{dt} \Big|_{t=t_0} \in T_{t_0} \mathbb{R} = T_{t_0}(-\varepsilon, \varepsilon)$. Also,

$$\gamma_{*,t_0} : T_{t_0}(-\varepsilon, \varepsilon) \rightarrow T_{\gamma(t_0)} M$$

Therefore $\gamma'(t_0) \in T_{\gamma(t_0)} M$.

Example 2.29. $M = \mathbb{R}^n$, $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^n$ via $t \mapsto (\gamma_1(t), \dots, \gamma_n(t)) = (x_1, \dots, x_n)$. Then,

$$\begin{aligned} \gamma'(t_0) &:= \gamma_{*,t_0} \left(\frac{d}{dt} \Big|_{t_0} \right) \\ &= \sum_{i=1}^n \underbrace{\gamma_{*,t_0} \left(\frac{d}{dt} \Big|_{t=t_0} \right) (x_i)}_{D\gamma_{t_0}} \frac{\partial}{\partial x_i} \Big|_{\gamma(t_0)} \end{aligned}$$

Therefore,

$$\gamma'(t_0) = \sum_{i=1}^n \gamma'_i(t_0) \frac{\partial}{\partial x_i} \Big|_{\gamma(t_0)}$$

So $(\gamma'_1(t_0), \dots, \gamma'_n(t_0))$ is the vector representation of $\gamma'(t_0)$ w.r.t $\{\frac{\partial}{\partial x_i} \Big|_{\gamma(t_0)}\}$. So, $\gamma'(t_0)$ can be interpreted as a vector in $\mathbb{R}^n$. In fact, any vector $v = (b_1, \dots, b_n)$ is the velocity of a curve in $\mathbb{R}^n$. For instance,

$$L : \gamma(t) = \gamma(t_0) + tv^{\rightarrow}, t \in \mathbb{R}$$

In general, on M , if $X_a \in T_a M$, then

$$X_a = \sum_{i=1}^n b_i \frac{\partial}{\partial x_i} \Big|_a$$

for some $b_1, \dots, b_n \in \mathbb{R}$. Let $\vec{v} = (b_1, \dots, b_n) \in \mathbb{R}^n$. Then, there exists a curve $\tilde{\gamma} : (-\varepsilon, \varepsilon) \rightarrow \varphi(U) \subset \mathbb{R}^n$ such that

$$\sum_{i=1}^n b_i \frac{\partial}{\partial x_i} \Big|_{\varphi(a)} = \tilde{\gamma}'(0)$$

Then,

$$\begin{aligned} X_a &= \sum_{i=1}^m b_i \frac{\partial}{\partial x_i} \Big|_a \\ &= (\varphi^{-1})_{*, \varphi(a)} \left(\sum_{i=1}^n b_i \frac{\partial}{\partial x_i} \Big|_{\varphi(a)} \right) \\ &= (\varphi^{-1})_{*, \varphi(a)} (\tilde{\gamma}'(0)) \\ &= (\varphi^{-1})_{*, \varphi(a)} \left(\tilde{\gamma}_{*, 0} \left(\frac{d}{dt} \Big|_{t=0} \right) \right) \end{aligned}$$

Therefore

$$X_a = (\varphi^{-1} \circ \tilde{\gamma})_{*, a} \left(\frac{d}{dt} \Big|_{t=0} \right) = \gamma_{*, 0} \left(\frac{d}{dt} \Big|_{t=0} \right) = \gamma'(0)$$

where $\gamma := \varphi^{-1} \circ \tilde{\gamma} : (-\varepsilon, \varepsilon) \rightarrow M$. Therefore, $X - a$ is the velocity of a curve in M with $\gamma(0) = a$. So,

$$T_a M = \{\text{velocity of curve on } M \text{ passing through } a\}.$$

Example 2.30. $M = M_{n \times n}(\mathbb{R})$ and $A \in M$. Then,

$$T_A M = \{H \in M_{n \times n}(\mathbb{R})\}$$

because $\forall H \in M_{n \times n}(\mathbb{R})$,

$$\gamma : \mathbb{R} \rightarrow M, t \mapsto A + tH$$

is a smooth curve in M with $A = \gamma(0)$, and

$$\gamma'(0) = \frac{d}{dt} \Big|_{t=0} (A + tH) = H$$

In particular, if $A = I_{n \times n}$, then $T_I M_{n \times n}(\mathbb{R}) = M_{n \times n}(\mathbb{R})$. Recall that $\text{GL}(n, \mathbb{R})$ is an open subset of $M_{n \times n}(\mathbb{R})$. Therefore,

$$T_A \text{GL}(n, \mathbb{R}) = T_A M_{n \times n}(\mathbb{R}), \forall A \in \text{GL}(n, \mathbb{R})$$

Definition 2.31. If G is a Lie group with identity e , then $T_e G$ is the Lie algebra of G .

Example 2.32. The Lie algebra of $\text{GL}(n, \mathbb{R})$ is

$$\mathfrak{gl}(n, \mathbb{R}) := T_I \text{GL}(n, \mathbb{R}) = M_{n \times n}(\mathbb{R})$$

Last(?) interpretation of $T_a M$:

Theorem 2.33. Let $F : M^n \rightarrow N^m$ smooth map and $c \in N$ such that $\forall a \in F^{-1}(c) \subset M$, DF_a is surjective. Then, $F^{-1}(c)$ is a smooth manifold of dimension $n - m$.

Proof. Apply result from $\tilde{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ after composing F with charts of M and N . □

Corollary 2.34. $\forall a \in F^{-1}(c)$, $T_a F^{-1}(c) = \ker(DF_a)$.

Proof. cute □

Example 2.35. $O(n) = \{A \in M_{n \times n}(\mathbb{R}) : AA^T = I\} = F^{-1}(I)$ with

$$F : M_{n \times n}(\mathbb{R}) \rightarrow S_{n \times n}(\mathbb{R}), A \mapsto AA^T$$

We can show that $DF_A(H) = AH^T + HA^T$, where $H = \gamma'(0)$, $\gamma(t) = A + tH$. Therefore

$$T_I O(n) = \ker(DF_I), \text{ where } DF_I(H) = H + H^T$$

$$\implies T_I O(n) = \{H \in M_{n \times n}(\mathbb{R}) : H^T = -H\} = \{\text{skew-symmetric matrices}\} = \mathfrak{o}(n).$$

3 Immersions, Embeddings, Submersions and Submanifolds

3.1 Immersions and embeddings and submersions

Let M, N are smooth manifolds of dimension m, n respectively, consider a smooth map $F : M \rightarrow N$. Then, $\forall p \in M$, $DF_p : T_p M \rightarrow T_{F(p)} N$ is an $\mathbb{R}$ -linear map.

Definition 3.1. 1. If DF_p is injective, then F is an immersion at p .

2. If $rm DF_p$ is surjective, then F is a submersion at p .

Also, if F is a immersion/submersion at every point on an open $U \subset M$, then F is an immersion/submersion on U .

Note:

1. If $U = M$, then F is simply called an immersion/submersion.
2. If F is an immersion at p , then $\dim T_p M \leq \dim T_{F(p)} N$. Since $\dim M = \dim T_p M$ and $\dim N = \dim T_{F(p)} N$, we have that $\dim M \leq \dim N$. Similarly, if F is a submersion at p , then

$$\dim T_p M \geq \dim T_{F(p)} N \implies \dim M \geq \dim N$$

Example 3.2. 1. Diffeomorphisms are both immersions and submersions on M because DF_p is an isomorphism $\forall p \in M$.

2. $M = \mathbb{R}^m$ and $N = \mathbb{R}^n$ with $m < n$. Then: $\iota : \mathbb{R}^m \rightarrow \mathbb{R}^n$, $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0, \dots, 0)$ is the canonical inclusion of $\mathbb{R}^m$ into $\mathbb{R}^n$ have

$$D\iota = \begin{pmatrix} I_m \\ 0 \end{pmatrix}$$

Then, $D\iota$ is injective if and only if $D\iota$ has rank $= \dim \mathbb{R}^m = m$. In this case, rank of $D\iota$ is $m \implies D\iota$ injective, for all $x \in \mathbb{R}^m$. Thus, ι is an immersion on $\mathbb{R}^m$.

3. $M = \mathbb{R}^m$ and $N = \mathbb{R}^n$ with $m > n$. Then, $\pi : \mathbb{R}^m \rightarrow \mathbb{R}^n$, $(x_1, \dots, x_n, x_{n+1}, \dots, x_m) \mapsto (x_1, \dots, x_n)$ is the canonical projection of $\mathbb{R}^m$ onto $\mathbb{R}^n$.

$$D\pi = [I_n \quad 0]$$

Then, $D\pi_x$ is surjective if and only if $D\pi_x$ has rank $= \dim(T_{\pi(x)}\mathbb{R}^n) = n$. It does, $\forall x$, so that π is a submersion on $\mathbb{R}^m$.

Theorem 3.3 (Canonical Immersion (Submersion) Theorem). Let $F : M \rightarrow N$ be an immersion (submersion) at $p \in M$. Then, $\dim M \leq (\geq) \dim N$ and $\exists$ charts $(U, \varphi), (V, \psi)$ of M, N respectively with $p \in U$ and $F(p) \in V$ such that $\psi \circ F \circ \varphi^{-1} = \iota|_{\varphi(U)}$ ($= \pi|_{\varphi(U)}$). These theorems follow from the Constant Rank Theorem

Definition 3.4. A smooth map $F : M \rightarrow N$ is said to have constant rank at $p \in M$ if DF_q has constant rank for any $q \in U$, where U is an open neighborhood of p .

Example 3.5. 1. Diffeomorphisms have constant rank on all M because DF_q is an isomorphism for all $q \in U$, U the open neighborhood of $p \in M$.

2. The canonical inclusion $\iota : \mathbb{R}^m \rightarrow \mathbb{R}^n, m < n$, has constant rank m on $\mathbb{R}^m$.

3. The canonical projection $\pi : \mathbb{R}^m \rightarrow \mathbb{R}^n, m > n$, has constant rank n on $\mathbb{R}^m$.

Note:

If $F : M \rightarrow N$ is an immersion (submersion) at p , then DF has constant rank $m = \dim M$ ($n = \dim N$) in an open neighborhood U (W) of p . So, F has constant rank at p .

Proof. Suppose that F is an immersion at p . Then,

$$DF_p : T_p M \rightarrow T_{F(p)} N$$

is injective. Let $(U, \varphi), (V, \psi)$ be charts of M, N respectively with $p \in U$ and $F(p) \in V$. Then,

$$\text{Jac}(\psi \circ F \circ \varphi^{-1})(\varphi(p))$$

is the matrix representation of DF_p with respect to the bases $\{\partial/\partial x_i|_p\}_{i=1}^m, \{\partial/\partial y_j|_{F(p)}\}_{j=1}^n$ of $T_p M$ and $T_{F(p)} N$ (where $\varphi = (x_1, \dots, x_m)$ and $\psi = (y_1, \dots, y_n)$). Since DF_p is injective, the rank of $\text{Jac}(\psi \circ F \circ \varphi^{-1})(\varphi(p))$ is equal to $\dim T_p M = m$. Thus, there exists $m \times m$ minor A of the matrix such that $\det A(\varphi(p)) \neq 0$. But $\det A$ is a smooth function on $\varphi(U)$. So, if $\det A(\varphi(p)) \neq 0$, then $\det A \neq 0$ on a neighborhood $\widetilde{W}$ of $\varphi(p)$ in $\varphi(U) \subset \mathbb{R}^m$. Set $W := \varphi^{-1}(\widetilde{W})$. Then, $p \in W$ and DF has rank m on W . $\square$

Theorem 3.6. Suppose that $F : M^m \rightarrow N^n$ is a smooth map that has constant rank r on an open neighborhood W of $p \in M$ so that $DF_q : T_q M \rightarrow T_{F(q)} N$ has rank r , $\forall q \in W$. Then, $\exists$ charts $(U, \varphi), (V, \psi)$ of M, N respectively with $p \in U$ and $F(p) \in V$ such that

$$\psi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V), (x_1, \dots, x_r, x_{r+1}, \dots, x_m) \mapsto (x_1, \dots, x_r, 0, \dots, 0)$$

where $\varphi = (x_1, \dots, x_m)$.

Note: $r \leq m$ since $\dim T_p M = m$

Proof. Let us first assume that $M \subset \mathbb{R}^m$ and $N \subset \mathbb{R}^n$ be open. So, $F : M \rightarrow N$, $x = (x_1, \dots, x_m) \mapsto (F_1(x), \dots, F_m(x))$, $DF = \left[\frac{\partial F_i}{\partial x_j} \right]_{\substack{i=1, \dots, n \\ j=1, \dots, m}}$ has rank r near p . So,

$$DF = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \dots & \frac{\partial F_1}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_r}{\partial x_1} & \frac{\partial F_r}{\partial x_2} & \dots & \frac{\partial F_r}{\partial x_m} \end{bmatrix}$$

After possibly permuting the variables $y_1, \dots, y_n$, we can assume that the first r rows are linearly independent near p . Therefore,

$$\begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \dots & \frac{\partial F_1}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_r}{\partial x_1} & \frac{\partial F_r}{\partial x_2} & \dots & \frac{\partial F_r}{\partial x_m} \end{bmatrix} \quad (1)$$

has rank r .

Set $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ by $x = (x_1, \dots, x_m) \mapsto (F_1(x), \dots, F_r(x), x_{r+1}, \dots, x_m)$. Then, $D\varphi$ has rank m , because the first r rows are equal to the matrix (1) and so linearly independent, and the bottom equal to $[O \ I_{m-r}]$. Therefore, by the Inverse Function Theorem, φ is locally invertible

with smooth inverse φ^{-1} . Set $F \circ \varphi^{-1}(x) = (\overbrace{x_1, \dots, x_r}^U, B(x_1, \dots, x_r))$ and $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $(y_1, \dots, y_r, y_{r+1}, y_n) \mapsto (y_1, \dots, y_r, V - B(U))$. Thus, $\psi \circ F \circ \varphi^{-1}(x) = (x_1, \dots, x_r, 0, \dots, 0)$ $\square$

Definition 3.7. Let M be a manifold of $\dim_{\mathbb{R}} = m$. Let $S \subset M$ be a subset such that $\forall p \in S$, there exists a chart (U, φ) of M with $p \in U$ such that

$$\varphi(U \cap S) = \{x \in \varphi(U) : x_{k+1} = \cdots = x_m = 0\}$$

for some fixed $k \leq m$. Then we say S is a k -dim smooth submanifold of M , with the topology on S is the subspace topology, and the smooth structure on S is inherited from M , that is, $\{(U \cap S, \varphi|_{U \cap S})\}$ with (U, φ) charts as above. Then,

$$i : S \rightarrow M \text{ via } p \mapsto p$$

is an embedding. Also, if $f : \tilde{S} \rightarrow M$ is any embedding, then $S := f(\tilde{S})$ is a submanifold of M . Also, for all $p \in S$, if (U, φ) is a chart from definition, then

$$T_p S := \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_k} \Big|_p \right\}$$

where $\varphi = (x_1, \dots, x_m)$.

Definition 3.8. A more general definition of submanifold is that of immersed submanifold. A subset $S \subset M$ is a k -dimensional immersed submanifold if $S = f(\tilde{S})$ with $\tilde{S}$ a k -dim manifold and $f : \tilde{S} \rightarrow M$ an immersion.

Example 3.9. Let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2, t \mapsto (t^2 - 1, t(t^2 - 1))$. Then,

$$S = \gamma(\mathbb{R}) = \{(x, y) = (t^2 - 1, t(t^2 - 1)), t \in \mathbb{R}\}$$

is an immersed 1-dim submanifold of $\mathbb{R}^2$ called the α -curve. This can never be an embedding. BUT, if f is not an embedding, then S is likely not a submanifold.

4 Smooth Vector Fields

Let M be a m -dim manifold. $\forall p \in M$

$$T_p M \cong \text{Der}_p(M)$$

$$= \{X_p : C^\infty(M) \rightarrow \mathbb{R} : X_p \text{ is } \mathbb{R}\text{-linear and } X_p(fg) = X_p(f)g(p) + f(p)X_p(g), \forall f, g \in C^\infty(M)\}$$

Also, if (U, φ) is a chart of M with $p \in U$ and $\varphi = (x_1, \dots, x_m)$, then

$$T_p M = \text{span}_{\mathbb{R}} \left\{ \frac{\partial}{\partial x_1} \Big|_p, \dots, \frac{\partial}{\partial x_m} \Big|_p \right\}$$

where

$$\frac{\partial}{\partial x_i} \Big|_p := (\varphi^{-1})_{*, \varphi(p)} \left(\frac{\partial}{\partial x_i} \Big|_{\varphi(p)} \right)$$

In particular, if $X_p \in T_p M$, then

$$X_p = \sum_{i=1}^m a_i \frac{\partial}{\partial x_i} \Big|_p$$

with $a_i = X_p(x_i) \in \mathbb{R}, \forall i = 1, \dots, m$. Therefore $(X_p(x_1), \dots, X_p(x_m))$ is a vector representation of X_p with respect to $\{\frac{\partial}{\partial x_i}|_p : i = 1, \dots, m\}$. Finally, if $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is an atlas of M with $\varphi_\alpha = (x_1^\alpha, \dots, x_m^\alpha)$, then $\forall p \in U_\alpha \cap U_\beta$,

$$\text{Jac}(\varphi_\alpha \circ \varphi_\beta^{-1})(p)$$

is the change of basis matrix from $\{\frac{\partial}{\partial x_i^\beta} : i = 1, \dots, m\}$ to $\{\frac{\partial}{\partial x_j^\alpha} : j = 1, \dots, m\}$. Let

$$TM := \bigsqcup_{p \in M} T_p M = \text{tangent bundle of } M$$

Then, TM is a manifold of $\dim 2m$: it admits the atlas $\{(V_\alpha, \Phi_\alpha)\}_{\alpha \in A}$ with

$$V_\alpha := \bigsqcup_{p \in U_\alpha} T_p M \subset TM$$

and

$$\begin{aligned} \Phi_\alpha : V_\alpha &\rightarrow \varphi_\alpha(U_\alpha) \times \mathbb{R}^m \subset \mathbb{R}^{2m} \text{ via} \\ T_p M \ni X_p &\mapsto (\varphi_\alpha(p), (X_p(x_1^\alpha), \dots, X_p(x_m^\alpha))) \in \mathbb{R}^m \times \mathbb{R}^m = \mathbb{R}^{2m} \end{aligned}$$

and

$$\Phi_\alpha \circ \Phi_\beta^{-1}(x, \vec{v}) = (\varphi_\alpha \circ \varphi_\beta^{-1}(x), \text{Jac}(\varphi_\alpha \circ \varphi_\beta^{-1})(x)\vec{v})$$

where $\vec{v} = (X_p(x_1^\beta), \dots, X_p(x_m^\beta))$ is the vector representations of X_p wrt $\{\frac{\partial}{\partial x_i^\beta}\}$ so that

$$\text{Jac}(\varphi_\alpha \circ \varphi_\beta^{-1})(x)\vec{v}$$

is the vector representation of X_p wrt $\{\frac{\partial}{\partial x_i^\alpha}\}$.

Definition 4.1. A (tangent) vector field on M is a map

$$X : M \rightarrow TM \text{ via } p \mapsto X(p) =: X_p \in T_p M$$

Let $\pi : TM \rightarrow M$ via $X_p \in T_p M \mapsto p$. Then, π is a submersion (Exercise!). Note that $X : M \rightarrow TM$ is a vector field if and only if

$$\pi \circ X = \text{id}_M$$

That is, X is a “section” of TM .

Definition 4.2. A vector field $X : M \rightarrow TM$ is called smooth if it is smooth as a map of manifolds.

NOTATION:

$$\mathfrak{X}(M) = \{\text{all smooth vector field on } M\}$$

Let $\{(U_\alpha, \varphi_\alpha = (x_1^\alpha, \dots, x_m^\alpha))\}_{\alpha \in A}$ be an atlas for M with corresponding atlas $\{(V_\alpha, \Phi_\alpha)\}_{\alpha \in A}$ of TM . Then, $\forall p \in U_\alpha$,

$$X(p) =: X_p \in T_p M = \text{span}_{\mathbb{R}}\{\frac{\partial}{\partial x_i^\alpha}|_p\}$$

$$\implies X_p = \sum_{i=1}^m X_p(x_i^\alpha) \frac{\partial}{\partial x_i^\alpha} |_p, X_p(x_i^\alpha) \in \mathbb{R}$$

Let $\alpha_i : U_\alpha \rightarrow \mathbb{R}$ via $p \mapsto X_p(x_i^\alpha)$, $\forall i = 1, \dots, m$. Then

$$X|_{U_\alpha} = \sum_{i=1}^m a_i \frac{\partial}{\partial x_i^\alpha}$$

where $a_i(p) = X(p)(x_i^\alpha)$, $\forall i = 1, \dots, m$. Then:

Proposition 4.3. X is smooth iff $X|_{U_\alpha}$ is smooth, $\forall \alpha$ iff the coefficient functions a_i are smooth, $\forall i$.

Proof. Exercise. □

Example 4.4. $M = \mathbb{R}^2, (x, y)$. Then

$$X = \cos x \frac{\partial}{\partial x} + ye^{x^2-1} \frac{\partial}{\partial y} \in \mathfrak{X}(\mathbb{R}^2)$$

Therefore $X(x, y) = (\cos x, ye^{x^2-1})$ is a basis $\{\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\}$.

Example 4.5. $X = y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y} \in \mathfrak{X}(\mathbb{R}^2)$.

4.1 Smooth Vector Fields as Derivations

Let $x \in \mathfrak{X}(M)$ so that $X(p) =: X_p \in T_p(M) = \text{Der}_p(M)$. Then, $X_p : C^\infty(M) \rightarrow \mathbb{R}$ is $\mathbb{R}$ -linear, and

$$X_p(fg) = X_p(f)g(p) + f(p)X_p(g), \forall f, g \in C^\infty(M)$$

Similarly, $X : C^\infty(M) \rightarrow C^\infty(M)$ via $f \mapsto X(f)$, where $X(f)(g) := X_p(f), \forall p \in M$, and X is $\mathbb{R}$ -linear, and

$$X(fg) = X(f)g + fX(g).$$

4.2 Smooth maps and vector fields

Let M, N be manifolds of dimension m, n respectively. Let $F : M \rightarrow N$ be a smooth map. Then $\forall p \in M$,

$$F_{*,p} : T_p M \rightarrow T_{F(p)} M \quad X_p \mapsto F_{*,p}(X_p)$$

where $F_{*,p}(X_p)(g) = X_p(g \circ F), \forall g \in C^\infty(N)$. Can we define a vector field F_*X on N such that

$$(F_*X)_{F(p)} = F_{*,p}(X_p), \forall p$$

This is not always possible! Suppose that F is not injective so that $F(p) = F(q)$ for $p \neq q$, and

$$F_{*,p}(X_p) \neq F_{*,q}(X_q)$$

Also, F may not be surjective so that F_*X may not be a vector field on all N .

But, if F is a diffeomorphism, then it is a bijection, and $F_{*,p}$ is an isomorphism $\forall p \in M$, so we can do it. This brings us to the following definition:

Definition 4.6. Let $X \in \mathfrak{X}(U)$ and $Y \in \mathfrak{X}(N)$. Then, X and Y are said to be F -related if

$$F_{*,p}(X(p)) = Y(F(p)), \forall p \in M$$

Proposition 4.7. X and Y are F -related if and only if

$$\forall f \in C^\infty(N), X(f \circ F) = (Y(f)) \circ F$$

Proof. By definition, X and Y are F -related

$$\begin{aligned} &\iff F_{*,p}(X(p)) = Y(F(p)), \forall p \in M \\ &\iff \forall f \in C^\infty(N), F_{*,p}(X(p))(f) = Y(F(p))(f) \end{aligned}$$

But

$$F_{*,p}(X(p))(f) = X(p)(f \circ F) = X(f \circ F)(p)$$

and

$$Y(F(p))(f) = Y(f)(F(p)) = Y(f) \circ F(p)$$

So $X \neq Y$ are F -related iff

$$X(f \circ F) = Y(f) \circ F, \forall f \in C^\infty(N).$$

□

Example 4.8. Let $F : \mathbb{R} = M \rightarrow \mathbb{R}^2 = N$ via $t \mapsto (\cos t, \sin t)$. Then

$$X = \frac{d}{dt} \text{ is } F\text{-related to } Y = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$$

Proof. Let $f \in C^\infty(\mathbb{R}^2)$, let us check that

$$X(f \circ F) = Y(f) \circ F$$

We have

$$\begin{aligned} X(f \circ F) &= X(f(\cos t, \sin t)) \\ &= \frac{d}{dt}(f(\cos t, \sin t)) \\ &= \frac{\partial f}{\partial x} \frac{d}{dt}(\cos t) + \frac{\partial f}{\partial y} \frac{d}{dt}(\sin t) \\ &= -\sin t \frac{\partial f}{\partial x}(\cos t, \sin t) + \cos t \frac{\partial f}{\partial y}(\cos t, \sin t), \end{aligned}$$

and

$$\begin{aligned} Y(f) \circ F(t) &= Y(f)(F(t)) \\ &= (-y \frac{\partial f}{\partial x} + x \frac{\partial f}{\partial y})(\cos t, \sin t) \\ &= -\sin t \frac{\partial f}{\partial x}(\cos t, \sin t) + \cos t \frac{\partial f}{\partial y}(\cos t, \sin t). \end{aligned}$$

Therefore $X(f \circ F) = Y(f) \circ F$.

□

Proposition 4.9. Let $X \in \mathfrak{X}(M)$. If F is a diffeomorphism, then there exists a unique vector field on N F -related to X , denoted F_*X , called the pushforward of X by F and given by:

$$(F_*X)_y = F_{*,F^{-1}(y)}(X(F^{-1}(y))), \forall y \in N.$$

Proof. Since F is a diffeomorphism, it is a bijection, so that $\forall y \in N$, there exists a unique $x \in M$ with $y = F(x)$. So,

$$(F_*X)_y = F_{*,F^{-1}(y)}(X(F^{-1}(y)))$$

is well-defined, $\forall y \in N$. Also, by definition, X and F_*X are F -related (since $x = F^{-1}(y)$). For uniqueness, assume that $Y \in \mathfrak{X}(N)$ is F -related to X . Then, $\forall y \in N$, if $x = F^{-1}(y)$,

$$\begin{aligned} Y(y) &= Y(F(x)) = F_{*,x}(X(x)) = F_{*,F^{-1}(y)}(X(F^{-1}(y))) = (F_*X)(y) \\ \implies Y &= F_*X \end{aligned}$$

□

Proposition 4.10. F_*X is a smooth vector field.

Proof. Exercise.

□

Proposition 4.11. Let $F : M \rightarrow N$ and $G : N \rightarrow Z$ be two diffeomorphisms, then

$$(F \circ G)_*X = F_*(G_*X), \forall X \in \mathfrak{X}(U)$$

Proof. Exercise.

□

Corollary 4.12. $\forall f \in C^\infty(N)$, $(F_*X)(f) \circ F = X(f \circ F)$, $\forall X \in \mathfrak{X}(M)$.

Proof. Follows from the fact that F_*X is F -related to X .

□

Corollary 4.13. $\forall X, X' \in \mathfrak{X}(M)$,

$$F_X[X, X'] = [F_*X, F_*X'].$$

Proof. Exercise.

□

How do you compute F_*X ? Suppose that $(U, \varphi = (x_1, \dots, x_m))$ is a chart for M . Then

$$X = \sum_{i=1}^m a_i \frac{\partial}{\partial x_i}$$

for some $a_i \in C^\infty(U)$. Also, if $F(U) \subset V$ where $(V, \psi = (y_1, \dots, y_n))$ is a chart on N , then

$$F_*X = \sum_{j=1}^m ((a_i \circ F^{-1}) \frac{\partial(y_j \circ F)}{\partial x_i}) \frac{\partial}{\partial y_j}.$$

Example 4.14. $M = \mathbb{R}^2, (x_1, x_2). N = \mathbb{R}^2, (y_1, y_2).$

$$X = x_1^2 \frac{\partial}{\partial x_1} - x_1(x_2 - 1) \frac{\partial}{\partial x_2}$$

$$F : \mathbb{R}^2 \rightarrow \mathbb{R}^2, (x_1, x_2) \mapsto (x_1 - x_2, x_2) = (y_1, y_2).$$

Then, $F^{-1}(y_1, y_2) = (y_1 + y_2, y_2)$. In the formula, $X = a_1 \frac{\partial}{\partial x_1} + a_2 \frac{\partial}{\partial x_2}$, where $a_1 = x_1^2, a_2 = -x_1(x_2 - 1)$, and

$$y_1 \circ F = x_1 - x_2, y_2 \circ F = x_2.$$

Then

$$\begin{aligned} a_1 \circ F^{-1}(y_1, y_2) &= (y_1 + y_2)^2 \\ a_2 \circ F^{-1}(y_1, y_2) &= -(y_1 + y_2)^2(y_2) - 1 \\ \frac{\partial(y_1 \circ F)}{\partial x_1} &= 1, \frac{\partial(y_1 \circ F)}{\partial x_2} = 1 \\ \frac{\partial(y_2 \circ F)}{\partial x_1} &= 0, \frac{\partial(y_2 \circ F)}{\partial x_2} = 1 \end{aligned}$$

$$\implies F_*X = [(y_1 + y_2)^2 + (y_1 + y_2)(x_2 - 1)] \frac{\partial}{\partial y_1} - (y_1 + y_2)(y_2 - 1) \frac{\partial}{\partial x_2}.$$

4.3 Integral curves and flows

Let $U \subset M$ be open, and $X \in \mathfrak{X}(U)$. Let $p \in U$.

Definition 4.15. An integral curve of X passing through p is the unique smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ such that

$$\gamma(0) = p, \text{ and } \dot{\gamma}(t) := \gamma'(t) = X(\gamma(t)), \forall t \in (-\varepsilon, \varepsilon).$$

Remark 4.16. The equation $\dot{\gamma}(t) = X(\gamma(t))$ corresponds to a system of ODE's with smooth coefficients and initial condition $=0$. Then the solution always exists and is unique.

Example 4.17. $M = \mathbb{R}^2, (x_1, x_2).$

1. $X = \frac{\partial}{\partial x_1}$. Let $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^2$ via $t \mapsto (x_1(t), x_2(t))$. Then

$$\dot{\gamma}(t) = (\dot{x}_1(t), \dot{x}_2(t)), X = 1 \frac{\partial}{\partial x_1} + 0 \frac{\partial}{\partial x_2}.$$

So,

$$\begin{aligned} \dot{\gamma}(t) = X(\gamma(t)) &\iff \begin{cases} \dot{x}_1(t) = 1 \\ \dot{x}_2(t) = 0 \end{cases} \\ &\iff \begin{cases} x_1 = t + c_1 \\ x_2 = c_2 \end{cases}, c_1, c_2 \in \mathbb{R} \end{aligned}$$

Therefore $\gamma(t) = (t + c_1, c_2) = t\vec{e}_1 + p$, where $p = (c_1, c_2)$, and $\vec{e}_1 = (1, 0)$. Similarly, if $Y = \frac{\partial}{\partial x_2}$, then its integral curves are $\gamma(t) = t\vec{e}_2 + p$.

2. $X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}$, $\gamma(t) = (x_1(t), x_2(t))$. Then

$$\begin{aligned}\dot{\gamma}(t) = X(\gamma(t)) &\iff \begin{cases} \dot{x}_1(t) = x_1(t) \\ \dot{x}_2(t) = x_2(t) \end{cases} \\ &\implies \begin{cases} x_1(t) = c_1 e^t \\ x_2(t) = c_2 e^t \end{cases} \\ &\implies \gamma(t) = e^t p, p = (c_1, c_2) \in \mathbb{R}^2\end{aligned}$$

3. $Y = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2}$, Note that at (x_1, x_2) , $Y_{(x_1, x_2)} = (-x_2, x_1)$. $\gamma(t) = (x_1(t), x_2(t))$. Then

$$\begin{aligned}\dot{\gamma}(t) = X(\gamma(t)) &\iff \begin{cases} \dot{x}_1(t) = -x_2(t) \\ \dot{x}_2(t) = x_1(t) \end{cases} \\ &\implies \begin{cases} x_1(t) = r_0 \cos(t + t_0) \\ x_2(t) = r_0 \sin(t + t_0) \end{cases}\end{aligned}$$

whith $\gamma(0) = (r_0 \cos t_0, r_0 \sin t_0) = p_0$, $\implies \gamma(t) = (r \cos t, r \sin t)$

Example 4.18. Let $U = \mathbb{R}^n$, $(x_1, \dots, x_m)$, let $\gamma(t) = (x_1(t), x_2(t), \dots, x_m(t))$ whose derivative is $\dot{\gamma}(t) = (\dot{x}_1(t), \dots, \dot{x}_m(t))$, and $X(t) = \sum_{i=1}^m a_i(x) \frac{\partial}{\partial x_i} \Big|_x$

We solve get

$$\begin{cases} \dot{\gamma}(t) = X(\gamma(t)) \\ \gamma(0) = p \end{cases} \iff \begin{cases} \dot{x}_i(t) = a_i(\gamma(t)), \forall i = 1, \dots, m \end{cases}$$

Definition 4.19. The local flow of a vector field $X \in \mathfrak{X}(U)$ is a smooth function

$$\Phi^X : (-\varepsilon, \varepsilon) \times U \rightarrow M \text{ via } (t, p) \mapsto \gamma_p^X(t),$$

where γ_p^X is the integral curve of X thorough p .

Note: Such flows exist. See Tu's (section 14.3) or Lee (Chapter 9).

Example 4.20. Let $U = \mathbb{R}^2, (x_1, x_2)$.

1. Let $X_i = \frac{\partial}{\partial x_i}$, then for all $p, \gamma_p^X(t) = t\vec{e}_i + p, t \in \mathbb{R} \implies \Phi^X(t, p) = t\vec{e}_i + p$. e.g.

$$\Phi^{X_1}(t, p = (p_1, p_2)) = (t + p_1, p_2)$$

2. Let $X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}$, then for all $p, \gamma_p^X(t) = e^t p \implies \Phi^X(t, p) = e^t p = (e^t p_1, e^t p_2)$.

3. $X = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2}$, then for all $p = (r \cos \alpha, r \sin \alpha)$, $\gamma_p^X = (r \cos(t + \theta), r \sin(t + \theta))$, so $\Phi^X(t, p) = (r \cos(t + \theta), r \sin(t + \theta))$.

NOTATION: Let $X \in \mathfrak{X}(U)$ with local flow Φ^X ,

- **Fix** p_0 : $\gamma_{p_0}^X(t) := \Phi^X(t, p_0)$ = (the integral curve of X through p_0).
- **Fix** t_0 : $\Phi_{t_0}^X(p) := \Phi^X(t_0, p) \in M$, where $\Phi_{t_0}^X : U \rightarrow M$ is the flow of X at time t_0 . E.g.
 1. $X_1 = \frac{\partial}{\partial x_1}$, then $\Phi^{X_1}(t, p) = te_1 + p$. Fix t_0 , then $\Phi_{t_0}^{X_1}(p) = p + t_0 e_1$
 2. $X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}$, $\Phi^X(t, p) = e^t p$. Fix t_0 , $\Phi_{t_0}^X(p) = e^{t_0} p$.

Let $X \in \mathfrak{X}(U)$ and suppose that $X(p) \neq 0, \forall p \in U$. Then, $\gamma_p^X(t)$ is a 1-dimensional sub-manifold of U because $\gamma_p^X : (-\varepsilon, \varepsilon) \rightarrow U$ such that $\dot{\gamma}_p^X(t) = X(\gamma_p^X(t)) \neq 0$. So, γ_p^X is an immersion which locally looks like $t \mapsto (t, 0, \dots, 0)$. Hence locally an embedding. So, around every $t_0 \in (-\varepsilon, \varepsilon)$, the image of γ_p^X is a 1-dimensional sub-manifold. And, $\dot{\gamma}_p^X(t)$ spans the tangent space of the sub-manifold.

Question: Let $X_1, \dots, X_k \in \mathfrak{X}(U)$ such that $\forall p \in U, \{X_1(p), \dots, X_k(p)\}$ is a linearly independent set of vectors in $T_p M$. Then, $D = \{X_1, \dots, X_k\}$ is called a smooth k -dist. of tangent vectors on U . Note that

$$\text{Span}_{\mathbb{R}}\{X_1(p), \dots, X_k(p)\} \subseteq T_p M$$

is the k -dimensional subspace.

Example 4.21. Let $M = \mathbb{R}^3, (x_1, x_2, x_3)$.

1. $D = \left\{ \frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2} \right\}$
2. $M = \mathbb{R}^2, (x_1, x_2), X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}, Y = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2}$, we have $D = \{X, Y\}$

Question: Does there exists a k -dimensional sub-manifold S of M such that $S \subset U$ and $T_p S = D, \forall p \in U$?

Answer: Yes, if $[X_i, X_j] = 0, \forall i, j = 1, \dots, k$.

FACT: Let $X, Y \in \mathfrak{X}(U)$. Then, $[X, Y] = 0$ if and only if $\Phi_t^X \circ \Phi_s^Y = \Phi_s^Y \circ \Phi_t^X$ for all $s, t \in (-\varepsilon, \varepsilon)$.

Proposition 4.22. Suppose that $\{X_1, \dots, X_k\}$ is a smooth distribution, such that $[X_i, X_j] = 0, \forall i, j = 1, \dots, k$. Fix $p_0 \in U$. Then,

$$\Psi : \mathbb{R}^k \rightarrow M \text{ via } (t_1, \dots, t_k) \mapsto \Phi_{t_1}^{X_1} \circ \dots \circ \Phi_{t_k}^{X_k}$$

is an embedding in a neighbourhood of 0 and

$$\frac{\partial \Psi}{\partial t_i} = X_i, \forall i.$$

Corollary 4.23. $S = \Psi(U_0)$ is a sub-manifold of M contained in U such that

$$T_p S = D_p.$$

Example 4.24. Let $M = \mathbb{R}^2, X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}, Y = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2}$, and $\Phi_t^X(p) = e^t p, \Phi_s^Y(p) = (r \cos(s + \theta), r \sin(s + \theta))$ where $p = (r \cos \theta, r \sin \theta)$. Then, $[X, Y] = 0$. And, our $\Psi(t_1, t_2) = (e^{t_1} \cos t_2, e^{t_1} \sin t_2)$.

Theorem 4.25 (Frobenius). Suppose that $D = \{X_1, \dots, X_k\}$ is a smooth distribution such that

$$[X_i, X_j](p) \in D(p). \quad (\star)$$

Then, you can find linearly independent smooth vectors fields $Y_1, \dots, Y_k$ such that $[Y_i, Y_j] = 0, \forall i, j = 1, \dots, k$, and

$$\text{Span}_{\mathbb{R}}\{Y_1(p), \dots, Y_k(p)\} = D_p.$$

Hence, there exists submanifold S of M with $T_p S = D_p$ if and only if $(\star)$ holds. In particular,

$$T_p S = \text{Span} \left\{ \left. \frac{\partial}{\partial x_1} \right|_p, \dots, \left. \frac{\partial}{\partial x_k} \right|_p \right\}, \text{ where } \left[\left. \frac{\partial}{\partial x_1} \right|_p, \left. \frac{\partial}{\partial x_1} \right|_p \right] = 0.$$

5 Tensors and Differential Forms

5.1 Tensors

Let V be a real vector spaces of dimension n , $\beta = \{e_1, \dots, e_n\}$ a basis of V . Then,

$$V^* := \{f : V \rightarrow \mathbb{R}, f \text{ linear}\}$$

1. $\forall f \in V^*, f$ is completely determined by

$$b_i := f(e_i), i = 1, \dots, n$$

Then, $\forall v \in V$, if $v = \sum_{i=1}^n a_i e_i$, then

$$f(v) = \sum_{i=1}^n a_i f(e_i) = \sum_{i=1}^n a_i b_i = [b_1 \quad \dots \quad b_n] \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

Therefore $[b_1 \quad \dots \quad b_n]$ is the matrix representation of f wrt β .

2. V^* is itself a real vector space of dim n with dual basis $\{e_i^*, \dots, e_n^*\}$, where

$$e_i^* : V \rightarrow \mathbb{R}, e_i^*(e_j) = \delta_{ij}$$

Then for all $f \in V^*, f = \sum_{i=1}^n b_i e_i^*$ (if $f(e_i) = b_i$).

Tensors are multilinear maps. For all $k \geq 1$, set

$$V^k = V \times \dots \times V = \{(v_1, \dots, v_n) : v_i \in V\}$$

Definition 5.1. A map $f : V^k \rightarrow \mathbb{R}$ is called

1. linear in the i -th variable if

$$f(v_1, \dots, v_i + \alpha w_i, \dots, v_k) = f(v_1, \dots, v_i, \dots, v_k) + \alpha f(v_1, \dots, w_i, \dots, v_k), \forall \alpha \in \mathbb{R}, v_1, \dots, v_n, w_i \in V$$

2. multilinear if it is linear in every i .

A k -tensor (a tensor of order k) is a multilinear map $f : V^k \rightarrow \mathbb{R}$. We denmote $\mathcal{L}^k(V)$ the space of all tensors.

Example 5.2. 1. If $k = 1$, then 1-tensors are just linear maps $f : V \rightarrow \mathbb{R}$, so $\mathcal{L}^1(V) = V^*$.

2. If $k = 2$, then 2-tensors are bilinear maps $f : V \times V \rightarrow \mathbb{R}$.

Recall: $f : V \times V \rightarrow \mathbb{R}$ bilinear is completely determined by

$$f(e_i, e_j) = a_{ij} \in \mathbb{R}$$

$$\text{So } \forall v = \sum_{i=1}^n v_i e_i, w = \sum_{j=1}^n w_j e_j,$$

$$\begin{aligned} f(v, w) &= f\left(\sum_{i=1}^n v_i e_i, \sum_{j=1}^n w_j e_j\right) \\ &= \sum_{i,j=1}^n v_i w_j f(e_i, e_j) \\ &= \sum_{i,j=1}^n v_i a_{ij} w_j \\ &= \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \underbrace{\begin{matrix} A \\ = (a_{ij}) \end{matrix}}_{= (a_{ij})} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \\ &= v^T A w \end{aligned}$$

So $f(v, w) = v^T A w$, with A is the matrix representation of f wrt β .

Remark 5.3. 1. If f is symmetric, so that $f(v, w) = f(w, v), \forall v, w \in V$. We have that $A^T = A$, that is, A is symmetric.

2. If f is skew-symmetric, so that $f(v, w) = -f(w, v), \forall v, w \in V$. We have that $A^T = -A$, that is, A is skew-symmetric.

3. For any $A \in M_{n \times n}(\mathbb{R})$,

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew-symmetric}}$$

If f is the 2-tensor corresponding to A , then

$$f = f^s + f^a$$

where f^s is the 2-tensor corresponding to $\frac{1}{2}(A + A^T)$, the symmetric part; f^a is the 2-tensor corresponding to $\frac{1}{2}(A - A^T)$, the skew-symmetric part.

Theorem 5.4. $\mathcal{L}^k(V)$ is a vector space with addition

$$(f + g)(v_1, \dots, v_k) = f(v_1, \dots, v_k) + g(v_1, \dots, v_k),$$

and scalar multiplication

$$(\alpha f)(v_1, \dots, v_k) = \alpha f(v_1, \dots, v_k)$$

Proof. Exercise. □

Remark 5.5. $\dim_{\mathbb{R}} \mathcal{L}^k(V) = n^k$ as a vector space.

Let us describe a basis for $\mathcal{L}^k(V)$. For any k -tuple $I = (i_1, \dots, i_k)$ of integers $i_j \in [n]$. We set

$$e_I := (e_{i_1}, \dots, e_{i_k})$$

Then I is called a multi-index. Note that the e_I 's do not span V^k because they cannot express $(e_i, 0, \dots, 0)$. But we have

Proposition 5.6. If $f, g \in \mathcal{L}^k(V)$ are such that $f(e_I) = g(e_I)$ for any multi-index I , then $f = g$.

Proof. Let $(v_1, \dots, v_k) \in V^k$. Then,

$$v_i = \sum_{j_i=1}^n c_{i,j_i} e_{j_i}$$

Then,

$$\begin{aligned} f(v_1, \dots, v_k) &= f\left(\sum_{j_1=1}^n c_{1,j_1} e_{j_1}, \dots\right) = \dots = g(v_1, \dots, v_k) \\ &\implies f = g \end{aligned}$$

□

For any multi-index $I = (i_1, \dots, i_k)$ with $i_1, \dots, i_k \in [n]$,

$$\Phi_I : V^k \rightarrow \mathbb{R} \text{ via } e_I \mapsto \begin{cases} 1, I = J \\ 0, I \neq J \end{cases}$$

and extend multi-linearly. One can check that

$$\Phi_I(v_1, \dots, v_k) = e_{i_1}^*(v_1) \dots e_{i_k}^*(v_k).$$

Theorem 5.7. $\{\Phi_I : I \in [n]^k\}$ is a basis of $\mathcal{L}^k(V)$ so that $\dim_{\mathbb{R}}(\mathcal{L}^k(V)) = n^k$.

Proof. Exercise. □

Definition 5.8. Let $f \in \mathcal{L}^k(V)$ and $g \in \mathcal{L}^l(V)$ with $k, l \geq 1$. We define the $(k + l)$ -tensor of f and g by

$$\begin{aligned} f \otimes g : V^{k+l} &\rightarrow \mathbb{R} \\ (v_1, \dots, v_{k+l}) &\mapsto f(v_1, \dots, v_k) g(v_{k+1}, \dots, v_{k+l}) \end{aligned}$$

Then, $f \otimes g \in \mathcal{L}^{k+l}(V)$ and satisfies

Proposition 5.9. Let f, g, h be tensors on V . Then:

1. $f \otimes (g \otimes h) = (f \otimes g) \otimes h$,
2. $(cf) \otimes g = c(f \otimes g) = f \otimes (cg), \forall c \in \mathbb{R}$,
3. $(f + g) \otimes h = f \otimes h + g \otimes h$ if $f, g \in \mathcal{L}^k(V)$, $f \otimes (g + h) = f \otimes g + f \otimes h$, if $g, h \in \mathcal{L}^k(V)$,
4. $\Phi_I = e_{i_1}^* \otimes \cdots \otimes e_{i_k}^*, \forall I = (i_1, \dots, i_k)$. So: $\Phi_I(v_1, \dots, v_k) = e_{i_1}^*(v_1) \cdots e_{i_k}^*(v_k)$

Definition 5.10. Let $f \in \mathcal{L}^k(V)$. Then, f is symmetric if

$$f(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = f(v_1, \dots, v_j, \dots, v_i, \dots, v_k), \forall i, j \in [n]$$

Also, f is skew-symmetric or alternating if

$$f(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = -f(v_1, \dots, v_j, \dots, v_i, \dots, v_k), \forall i, j \in [n]$$

Example 5.11. 1. Inner products are symmetric bilinear maps, so it is a symmetric 2-tensors.

2. Let $\det : V^n \rightarrow \mathbb{R}$ via

$$(v_1, \dots, v_n) \mapsto \det [v_1 \mid \cdots \mid v_n]$$

Then, $\det$ is skew-symmetric.

Let $A^k(V) = \{\text{all alternating } k\text{-tensors}\} \subset \mathcal{L}^k(V)$ a subspace. If $k = 1$, then $A^1(V) = \mathcal{L}^1(V)$. But $A^k(V) \subsetneq \mathcal{L}^k(V), k > 1$.

Proposition 5.12. Let $f \in A^k(V)$. Then:

1. $f(v_1, \dots, v_k) = 0$ if $v_p = v_q$ for some $p \neq q$.
2. f is completely determined by $f(e_I)$, where $I = \{i_1, \dots, i_k\}$ with $i_1 < \cdots < i_k$.

Proof. 1. Suppose $v_p = v_q$ with $p < q$. Then

$$f(v_1, \dots, v_p, \dots, v_q, \dots, v_k) = -f(v_1, \dots, v_q, \dots, v_p, \dots, v_k) = -f(v_1, \dots, v_p, \dots, v_q, \dots, v_k)$$

$$\text{So } f(v_1, \dots, v_p, \dots, v_q, \dots, v_k) = 0.$$

2. If $I = (i_1, \dots, i_k)$ is such that $i_l = i_{l'}$ for some $1 \leq l < l' \leq k$, then $f(e_I) = f(e_{i_1}, \dots, e_{i_l}, \dots, e_{i_{l'}}, \dots, e_{i_k}) = 0$ by (1). So we only need to keep track of distinct i_l 's. Also, by skew-symmetric,

$$f(e_I) = f(e_{i_1}, \dots, e_{i_l}, \dots, e_{i_{l'}}, \dots, e_{i_k}) = -f(e_I) = f(e_{i_1}, \dots, e_{i_{l'}}, \dots, e_{i_l}, \dots, e_{i_k})$$

So we only need to keep track of the increasing i_l 's.

□

Theorem 5.13. Φ_I with $I = (i_1, \dots, i_k), i_1 < \cdots < i_k$ is a basis of $A^k(V)$.

Example 5.14. $f \in A^2(V)$, then

$$f = a_{11}e_1^* \otimes e_1^* + a_{12}e_1^* \otimes e_2^* + a_{21}e_2^* \otimes e_1^* + a_{22}e_2^* \otimes e_2^*$$

where $a_{ij} = f(e_i, e_j) \in \mathbb{R}$. Also,

$$\begin{aligned} a_{11} &= f(e_1, e_1) = 0 \\ a_{22} &= f(e_2, e_2) = 0 \\ a_{12} &= f(e_1, e_2) = -f(e_2, e_1) = -a_{21} \\ \implies f &= a_{12}e_1^* \otimes e_2^* - a_{12}e_2^* \otimes e_1^* = a_{12}(e_1^* \otimes e_2^* - e_2^* \otimes e_1^*) \end{aligned}$$

We set

$$e_1^* \wedge e_2^* := e_1^* \otimes e_2^* - e_2^* \otimes e_1^* = \sum_{\sigma \in S_2} (\text{sgn} \sigma) e_{\sigma(1)}^* \otimes e_{\sigma(2)}^*$$

Alternatively,

$$e_1^* \wedge e_2^* = \sum_{\sigma \in S_2} (\text{sgn} \sigma) e_1^*(v_{\sigma(1)}) e_2^*(v_{\sigma(2)})$$

So, $\Psi_I := e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$ with $i_1 < \cdots < i_k$ is defined as

$$\Psi_I(v_1, \dots, v_k) := \sum_{\sigma \in S_k} (\text{sgn} \sigma) e_{i_1}^*(v_{\sigma(1)}) \cdots e_{i_k}^*(v_{\sigma(k)})$$

Then, $\Psi_I \in A^k(V)$ and $\{\Psi_I : I = (i_1, \dots, i_k), i_1 < \cdots < i_k\}$ is a basis of $A^k(V)$.

$$\implies A^k(V) \text{ is an } \mathbb{R}\text{-vector space of dimension } \binom{n}{k}.$$

CONVENTION: $A^0(V) = \mathbb{R}$. Notice that

$$A^n(V) = \mathbb{R} = \{c \det : c \in \mathbb{R}\}, \text{ where } \det : V^n \rightarrow \mathbb{R}, (v_1, \dots, v_n) \mapsto \det [v_1 \mid \cdots \mid v_n]$$

as $A^n(V)$ has dimension $\binom{n}{n} = 1$. Also, $A^k(V) = \{0\}$ if $k > n$.

Now let's introduce the general formula for wedge products.

Definition 5.15. Let $f \in A^k(V), g \in A^l(V)$. Set

$$f \wedge g(v_1, \dots, v_{k+l}) = \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} (\text{sgn} \sigma) f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) g(v_{\sigma(k+1)}, \dots, v_{\sigma(k+l)})$$

Proposition 5.16. If f, g, h are alternating tensors, then:

1. $f \wedge (g \wedge h) = (f \wedge g) \wedge h$
2. $(cf) \wedge g = c(f \wedge g) = f \wedge (cg), \forall c \in \mathbb{R}$
3. $(f + g) \wedge h = f \wedge h + g \wedge h$
4. If $f \in A^k(V)$ and $g \in A^l(V)$, then

$$f \wedge g = (-1)^{kl} g \wedge f$$

Example 5.17. $e_1^* \otimes e_2^* = (e_1^* \otimes e_2^* - e_2^* \otimes e_1^*) = -(e_2^* \otimes e_1^* - e_1^* \otimes e_2^*) = (-1)^{1 \cdot 1} e_2^* \wedge e_1^*$

NOTATION: $\bigwedge^k V^* := A^k(V)$, spanned by $e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$ with $e_{i_j}^* \in V^*$.

5.2 Differential Forms

M is an n -dim manifold. Let $(U, \varphi = (x_1, \dots, x_n))$ be a chart. Then, $\forall p \in U$, $T_p M = \text{span}_{\mathbb{R}} \{ \frac{\partial}{\partial x_i} |_p \}$

$$T_p^* M = (T_p M)^* = \text{span}_{\mathbb{R}} \{ (dx_i)_p \}$$

where $(dx_i)_p(\frac{\partial}{\partial x_j} |_p) = \delta_{ij}$.

$$T^* M = \bigsqcup_{p \in M} T_p^* M = \text{cotangent bundle, a } 2n\text{-dim manifold}$$

If $T^* M|_U := \bigsqcup_{p \in U} T_p^* M$, then

$$\Phi : T^* M|_U \rightarrow \varphi(U) \times \mathbb{R}^n \subset \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n}$$

via

$$\sum_{i=1}^n b_i(p)(dx_i)_p =: \omega_p \in T_p^* M \mapsto (\varphi(p), (b_1(p), \dots, b_n(p)))$$

Definition 5.18. A smooth differential 1-form on M is a smooth map $\omega : M \rightarrow T^* M$ such that

$$\omega(p) \in T_p^* M, \forall p \in M$$

In general, $\bigwedge^k T^* M := \bigsqcup_{p \in M} \bigwedge^k T_p^* M$, where $\forall p \in M$,

$$\bigwedge^k T_p^* M = \text{span}_{\mathbb{R}} \{ (dx_{i_1})_p \wedge \dots \wedge (dx_{i_k})_p : i_1 < \dots < i_k \}$$

We get charts

$$\bigwedge^k \Phi : \bigwedge^k T^* M|_U = \bigsqcup_{p \in U} \bigwedge^k T_p^* M \rightarrow \varphi(U) \times \mathbb{R}^{\binom{n}{k}}$$

via

$$\alpha_p = \sum_I a_I(p)(dx_{i_1})_p \wedge \dots \wedge (dx_{i_k})_p \mapsto (\varphi(p), (a_I(p)))$$

Definition 5.19. A smooth differential k -form on M is a smooth map

$$\omega : M \rightarrow \bigwedge^k T^* M$$

such that $\omega(p) \in \bigwedge^k T_p^* M$.

CONVENTION: $\bigwedge^0 T^* M := C^\infty(M)$, as we convent each $\bigwedge^0 T_p^* M = \mathbb{R}$.

NOTATION: $\bigwedge^k M = \{(\text{smooth}) \text{ differential } k\text{-forms}\}$.

We would like to look at the differential 1-forms first. Let $\omega \in \bigwedge^1 M$ such that

$$\omega : M \rightarrow T^* M$$

via

$$p \mapsto \omega(p) \in T_p^*M$$

In local coordinates $(U, \varphi = (x_1, \dots, x_n))$,

$$\omega(p) = \sum_{i=1}^n a_i(p)(dx_i)_p$$

NOTATION: $\omega_p := \omega(p)$, $d_px_i := (dx_i)_p$. Then we have

$$\omega_p = \sum_{i=1}^n a_i(p)d_px_i$$

So for $a_i : U \rightarrow \mathbb{R}$,

$$\omega|_U] \sum_{i=1}^n a_i dx_i$$

BUT, $\omega : M \rightarrow T^*M$ is smooth. So,

$$\Phi \circ \omega \circ \varphi^{-1} : \varphi(U) \rightarrow \varphi(U) \times \mathbb{R}^n$$

$$x \mapsto \varphi^{-1}(x) \mapsto \omega(\varphi^{-1}(x)) \mapsto (\varphi(\varphi^{-1}(x)), (a_1 \circ \varphi^{-1}(x), \dots, a_n \circ \varphi^{-1}(x))) \text{ is smooth}$$

$$\iff a_i \circ \varphi^{-1} : \varphi(U) \subset \mathbb{R}^n \rightarrow \mathbb{R} \text{ are smooth, } \forall i$$

$$\iff a_i \in C^\infty(U)$$

Proposition 5.20. ω is smooth iff locally, $\omega = \sum_{i=1}^n a_i dx_i$ with $a_i \in C^\infty(U)$.

$$X \in \mathfrak{X}(U) \rightsquigarrow X : C^\infty(M) \rightarrow C^\infty M, \text{ } \mathbb{R}\text{-linear and Leibniz}$$

Can also interpret 1-form as maps:

$$\omega : \mathfrak{X}(M) \rightarrow C^\infty(M), X \mapsto \omega(X)$$

where

$$\omega(X)(p) = \omega_p(X_p) \in \mathbb{R}$$

with $\omega_p = \omega(p)$, $X_p = X(p)$. In local coordinates:

$$\omega = \sum_{i=1}^n a_i dx_i \text{ and } X = \sum_{j=1}^n b_j \frac{\partial}{\partial x_j}$$

$$\begin{aligned}
\implies \omega_p(X_p) &= \sum_{i=1}^n a_i(p) d_p x_i \left(\sum_{j=1}^n b_j(p) \frac{\partial}{\partial x_j} \right) \\
&= \sum_{i,j=1}^n a_i(p) b_j(p) \underbrace{d_p(x_i) \frac{\partial}{\partial x_j}}_{=\delta_{ij}} \\
&= \sum_{i=1}^n a_i(p) b_i(p)
\end{aligned}$$

$$\begin{aligned}
\implies \omega(X) &= \sum_{i=1}^n a_i b_i \text{ on } (U, \varphi = (x_1, \dots, x_n)) \text{ with } a_i, b_i \in C^\infty(U) \\
&\implies \omega(X) \in C^\infty(U)
\end{aligned}$$

NOTE: $\omega(fX) = f\omega(X), \forall f \in C^\infty(M)$. So

$$\omega : \mathfrak{X}(M) \rightarrow C^\infty(M) \text{ is } C^\infty(M)\text{-linear}$$

Recall that if we have a smooth map

$$F : M \rightarrow N$$

then for all $p \in M$,

$$F_{*,p} : T_p M \rightarrow T_{F(p)} N$$

Consider the dual map

$$(F_{*,p})^* : T_{F(p)}^* N \rightarrow T_p^* M$$

via

$$\omega_{F(p)} \mapsto (F_{*,p})^*(\omega_{F(p)})$$

where

$$(F_{*,p})^*(\omega_{F(p)}) : T_p M \rightarrow \mathbb{R}$$

via

$$X_p \mapsto \omega_{F(p)}(\underbrace{F_{*,p}(X_p)}_{\in T_{F(p)} N})$$

So given any $\omega \in \bigwedge^1 N$, we get a smooth 1-form $F^*\omega$ on M such that

$$(F^*\omega)_p := (F_{*,p})^*(\omega_{F(p)}), \forall p \in M$$

This form is called the pullback of ω along $F : M \rightarrow N$.

Change of Coordinates $(U, \varphi = (x_1, \dots, x_n)), (V, \psi = (y_1, \dots, y_n))$ charts on M . On $U \cap V$,

$$\begin{aligned}
(y_1, \dots, y_n) &= \psi \circ \varphi^{-1}(x_1, \dots, x_n) \\
\implies y_j &= y_j(x_1, \dots, x_n)
\end{aligned}$$

$$\frac{\partial}{\partial x_i} = \sum_{j=1}^n \frac{\partial y_j}{\partial x_i} \frac{\partial}{\partial y_j}, \forall i = 1, \dots, n$$

$$dy_j = \sum_{i=1}^n \frac{\partial y_j}{\partial x_i} dx_i, \forall j = 1, \dots, n$$

So

$$\omega|_U = \sum_{i=1}^n a_i dx_i, \quad \omega|_V = \sum_{j=1}^n b_j dy_j$$

On $U \cap V$:

$$\begin{aligned} \sum_{i=1}^n a_i dx_i &= \sum_{j=1}^n b_j dy_j \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n b_j \frac{\partial y_j}{\partial x_i} \right) dx_i \\ \implies a_i &= \sum_{j=1}^n b_j \frac{\partial y_j}{\partial x_i}, \forall i = 1, \dots, n \quad (*) \end{aligned}$$

So, a 1-form can be specified by taking scalar functions $a_1, \dots, a_n$ on U , and $b_1, \dots, b_n$ on V that satisfy (*). Note that let $f \in C^\infty(M)$. Then, $\forall p \in U$,

$$(df)_p := \sum_{i=1}^n \frac{\partial f}{\partial x_i}(p) (dx_i)_p \in T_p^* M$$

Note that $(df)_p$ is independent of the coordinates $(x_1, \dots, x_n)$ because if $p \in U \cap V$,

$$\begin{aligned} (df)_p &= \sum_{j=1}^n \frac{\partial f}{\partial y_j}(p) (dy_j)_p \\ &= \sum_{i=1}^n \frac{\partial y_j}{\partial x_i}(p) (dx_i)_p \\ &= \sum_{i=1}^n \left(\sum_{j=1}^n \frac{\partial f}{\partial y_j}(p) \frac{\partial y_j}{\partial x_i}(p) \right) (dx_i)_p \\ &= \sum_{i=1}^n \frac{\partial f}{\partial x_i}(p) (dx_i)_p \end{aligned}$$

$\rightsquigarrow$ get a well-defined global 1-form such that

$$df|_U = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

Since $\frac{\partial f}{\partial x_i}$ is smooth $\forall i = 1, \dots, n$, $df|_n$ is smooth, $\forall U$, therefore df is smooth.

Example 5.21. $M = \mathbb{R}^2$, $f : M \rightarrow \mathbb{R}$, $(x, y, z) \mapsto xyz^3$

$$\implies df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = yz^3dx + xz^3dy + 3xyz^2dz$$

Pullback: $F : M \rightarrow N$ smooth map. The pullback map

$$F^* : \bigwedge^1 N \rightarrow \bigwedge^1 M \quad \omega \mapsto F^*\omega$$

where $(F^*\omega)(p)(X_p) := \omega(F(p))(F_{*,p}(X_p))$, $\forall p \in M, X_p \in T_pM$. In local coordinates, if $(V, \psi = (y_1, \dots, y_m))$ are coordinates on N and $\omega|_V = \sum_{j=1}^n a_j dy_j$, then

$$F^*\omega = \sum_{j=1}^n (a_j \circ F) \left(\sum_{i=1}^n \frac{\partial (y_j \circ F)}{\partial x_i} dx_i \right)$$

$$\iff F^*\omega = \sum_{j=1}^n (a_j \circ F) d(y_j \circ F)$$

Example 5.22. $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ via $(x_1, x_2, x_3) \mapsto (2x_1 + x_2, x_3^2) = (y_1, y_2)$. Let $\omega = e^{y_1} dy_1 + y_1 y_2 dy_2$. Then,

$$\begin{aligned} F^*\omega &= e^{2x_1+x_2} d(2x_1 + x_2) + (2x_1 + x_2) x_3^2 d(x_3^2) \\ &= e^{2x_1+x_2} [2dx_1 + dx_2] + (2x_1 + x_2) x_3^2 (2x_3 dx_3) \end{aligned}$$

k-forms: $\omega : M \rightarrow \wedge^k T^*M = \bigsqcup \wedge^k T_p^*M$. In local coordinates $(U, \varphi = (x_1, \dots, x_n))$, then $\forall p \in U$,

$$\begin{aligned} \bigwedge^k T_p^*M &= A^k(T_pM) = \{\omega(p) : T_pM \times \dots \times T_pM \rightarrow \mathbb{R} : \omega(p) \text{ is alternating multilinear}\} \\ &= \text{span}_{\mathbb{R}}\{(dx_{i_1})_p \wedge \dots \wedge (dx_{i_k})_p : i_1 < \dots < i_k\} \\ &\rightsquigarrow \omega(p) = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k}(p) (dx_{i_1})_p \wedge \dots \wedge (dx_{i_k})_p \text{ with } a_{i_1, \dots, i_k}(p) \in \mathbb{R} \\ &\rightsquigarrow \omega|_U = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k} dx_{i_1} \wedge \dots \wedge dx_{i_k} \text{ and } \omega|_U \text{ is smooth iff } a_{i_1, \dots, i_k} \text{'s are smooth} \end{aligned}$$

Also, $\omega : \mathfrak{X}(M) \times \dots \times \mathfrak{X}(M) \rightarrow C^\infty(M)$ via $(X_1, \dots, X_k) \mapsto \omega(X_1, \dots, X_k)$, where

$$\omega(X_1, \dots, X_k)(p) := \omega(p)(X_1(p), \dots, X_k(p)), \forall p \in U$$

Example 5.23. $\mathbb{R}^3, (x, y, z)$,

$$\begin{aligned} \omega &= dx \wedge dy \rightsquigarrow \text{smooth 2-form} \\ \omega' &= (x^2y - 3z)dx \wedge dy \wedge dz \rightsquigarrow \text{smooth 3-form.} \end{aligned}$$

n -form: $\omega|_U = adx_1 \wedge \cdots \wedge dx_n$, with $a \in C^\infty(U)$, $\omega|_V = bdy_1 \wedge \cdots \wedge dy_n$, with $b \in C^\infty(M)$.
On $U \cap V$:

$$\begin{aligned} dx_1 \wedge \cdots \wedge dx_n &= \left(\sum_{j=1}^n \frac{\partial x_1}{\partial y_j} dy_j \right) \wedge \cdots \wedge \left(\sum_{j=1}^n \frac{\partial x_n}{\partial y_j} dy_j \right) \\ &= \det \left(\frac{\partial x_i}{\partial y_j} \right) dy_1 \wedge \cdots \wedge dy_n \\ &\rightsquigarrow adx_1 \wedge \cdots \wedge dx_n = a \det \left(\frac{\partial x_i}{\partial y_j} \right) dy_1 \wedge \cdots \wedge dy_n = bdy_1 \wedge \cdots \wedge dy_n \\ &\implies a \det \left(\frac{\partial x_i}{\partial y_j} \right) = b. \end{aligned}$$

Operations on k -forms.

Pullback: If $F : M \rightarrow N$, we define

$$F^* : \bigwedge^k N \rightarrow \bigwedge^k M \quad \omega \mapsto F^* \omega$$

where for all $p \in M$,

$$(F^* \omega)(p)(v_1, \dots, v_k) := \omega(F(p))(F_{*,p}(v_1), \dots, F_{*,p}(v_k)), \forall v_1, \dots, v_k \in T_p M$$

In local coordinates $(V, \psi = (y_1, \dots, y_n))$ on N , if

$$\omega|_U = \sum_{i_1 < \cdots < i_k} a_{i_1, \dots, i_k} dy_{i_1} \wedge \cdots \wedge dy_{i_k}$$

then,

$$F^* \omega = \sum_{i_1 < \cdots < i_k} (a_{i_1, \dots, i_k} \circ F) d(y_{i_1} \circ F) \wedge \cdots \wedge d(y_{i_k} \circ F)$$

Extension derivative $d : \bigwedge^k M \rightarrow \bigwedge^{k+1} M$:

* $k = 0$, $\bigwedge^0 M = C^\infty(M)$, $\forall f \in C^\infty(M)$,

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

* $k > 0$, $\forall \omega \in \bigwedge^k M$, if $\omega = \sum_{i_1 < \cdots < i_k} a_{i_1, \dots, i_k} dx_{i_1} \wedge \cdots \wedge dx_{i_k}$, then

$$d\omega = \sum_{i_1 < \cdots < i_k} (da_{i_1, \dots, i_k}) \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k}$$

Integration on M $M = \mathbb{R}^n$. Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is compactly supported (i.e., $\text{supp}(f)$ is closed and bounded), then

$$\int_{\mathbb{R}^n} f dx_1 \cdots dx_n = \int_{\text{supp}(f)} f dx_1 \cdots dx_n$$

which is computed using usual iterated integrals. Set $\omega = f dx_1 \wedge \cdots \wedge dx_n$ and

$$\int_{\mathbb{R}^n} \omega = \int_{\mathbb{R}^n} f dx_1 \cdots dx_n$$

Recall: change of variable $x_i = x_i(y_1, \dots, y_n)$

$$\implies dx_1 \cdots dx_n = |\det(\frac{\partial x_i}{\partial y_j})| dy_1 \cdots dy_n$$

But, $dx_1 \wedge \cdots \wedge dx_n = \det(\frac{\partial x_i}{\partial y_j}) dy_1 \wedge \cdots \wedge dy_n = |\det(\frac{\partial x_i}{\partial y_j})| dy_1 \wedge \cdots \wedge dy_n \iff \det(\frac{\partial x_i}{\partial y_j}) > 0$ on $U \cap V$.

Definition 5.24. A manifold M is called orientable if it admits an atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ such that

$$\det(Jac(\varphi_\alpha \circ \varphi_\beta^{-1})) > 0 \quad (**)$$

on $U_\alpha \cap U_\beta$.

Given an orientable manifold M with atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ satisfying (**), and $\omega \in \bigwedge^n M$, we define

$$\int_M \omega = \sum_{\alpha \in A} f_\alpha \int_{\varphi(U_\alpha)} (\varphi_\alpha^{-1})^*(\omega|_{U_\alpha})$$

where $\{f_\alpha : M \rightarrow \mathbb{R}\}_{\alpha \in A}$ is a partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$.

5.3 Stokes' Theorem

This is a generalization of the Fundamental Theorem of Calculus:

$$\int_a^b \frac{df}{dx} dx = f(b) - f(a)$$

For manifold, we have

$$\int_M d\omega = \int_{\partial M} \omega$$